

Forecasting USD/LKR, GBP/LKR and EUR/LKR Exchange Rates for Overseas Tuition Planning in Sri Lanka

CM3023 - Financial Time Series Analysis

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Chapter 1: Introduction

1.1 Background of the Study

International education represents a significant financial commitment for Sri Lankan students and their families. Students who pursue higher education overseas are generally required to pay tuition fees in foreign currencies, such as the United States dollar (USD), British pound sterling (GBP), and euro (EUR). However, families in Sri Lanka often need to arrange these payments using Sri Lankan rupees (LKR). Consequently, changes in exchange rates can have a considerable effect on the local-currency cost of overseas education.

Even when a university's tuition fee remains unchanged in its original currency, the amount required in Sri Lankan rupees can fluctuate due to movements in the exchange rate. When a foreign currency appreciates against the Sri Lankan rupee, the cost of paying the same tuition fee increases in LKR. Conversely, a depreciation of the foreign currency against the rupee can reduce the local-currency cost. These fluctuations introduce financial uncertainty and make it difficult for students and their families to estimate future education expenses accurately.

Exchange-rate forecasting provides a potential approach to managing this uncertainty. By analysing historical exchange-rate observations, statistical and machine-learning models can be developed to estimate future exchange rates and assess the uncertainty associated with those estimates. Short-term forecasts may help families anticipate upcoming payments, while longer forecasting horizons can provide additional information for financial planning.

Financial exchange-rate data are time-dependent and may exhibit characteristics such as changing volatility, autocorrelation, non-stationarity and sudden structural changes. These characteristics create challenges for forecasting and make it important to select suitable modelling techniques. Traditional time-series methods, including autoregressive integrated moving average (ARIMA) models, provide statistical approaches for modelling temporal relationships. Machine-learning methods, such as Random Forest and Extreme Gradient Boosting (XGBoost), offer alternative approaches that can capture nonlinear relationships in historical data. Volatility models, such as Generalised Autoregressive Conditional Heteroskedasticity (GARCH), can also be used to investigate changes in exchange-rate uncertainty.

This study examines exchange-rate forecasting for overseas tuition planning in Sri Lanka by comparing traditional time-series and machine-learning approaches. It focuses on forecasting exchange rates five and twenty reporting days ahead, analysing exchange-rate volatility, and translating forecast results into estimated tuition costs in Sri Lankan rupees. The study also considers prediction intervals to communicate the uncertainty surrounding future exchange rates and tuition expenses.

1.2 Problem Statement

Sri Lankan students planning to pursue higher education overseas face uncertainty regarding the future cost of tuition payments denominated in foreign currencies. Although the tuition fee specified by a foreign university may remain constant, fluctuations in the exchange rate can cause the equivalent cost in Sri Lankan rupees to increase or decrease. This uncertainty can complicate budgeting and financial preparation for students and their families.

Historical exchange-rate data provide information about past movements and statistical patterns. However, the extent to which these patterns can be used to forecast future exchange rates is uncertain. Traditional time-series and machine-learning models may perform differently depending on the currency, forecasting horizon, volatility conditions and underlying data characteristics.

Therefore, the central problem addressed by this study is to determine how effectively different forecasting approaches can predict future exchange rates and how the resulting forecasts can be used to estimate overseas tuition costs in LKR. In addition to point forecasts, the study considers forecast uncertainty to provide a more informative assessment of potential future payment requirements.

1.3 Research Question

The main research question of this study is:

How effectively can traditional time-series and machine-learning models forecast USD/LKR, GBP/LKR and EUR/LKR exchange rates five and twenty reporting days ahead, and how can these forecasts and their associated uncertainty support overseas tuition planning for Sri Lankan students and their families?

The study also addresses the following supporting questions:

1. What statistical characteristics, including stationarity, autocorrelation and volatility, are present in the selected exchange-rate series?
2. How do traditional time-series models compare with machine-learning models and a baseline forecasting method?
3. Does forecasting performance differ across currencies and between the five-day and twenty-day forecasting horizons?
4. How effectively can volatility models estimate future exchange-rate uncertainty?
5. How can the forecasts and prediction intervals be translated into estimated tuition costs in Sri Lankan rupees?

1.4 Research Objectives

1.4.1 Main Objective

The main objective of this study is to evaluate and compare traditional time-series and machine-learning methods for forecasting selected foreign exchange rates relevant to overseas tuition planning in Sri Lanka and to assess the practical usefulness of the forecasts for estimating future tuition costs and associated uncertainty.

1.4.2 Specific Objectives

The specific objectives are as follows:

1. To analyse historical USD/LKR, GBP/LKR and EUR/LKR exchange-rate data using exploratory data analysis and appropriate statistical techniques.
2. To investigate the stationarity, autocorrelation, distributional characteristics and volatility of the selected exchange-rate series.
3. To develop exchange-rate forecasting models for five-reporting-day and twenty-reporting-day horizons using traditional time-series and machine-learning approaches.
4. To compare the performance of the forecasting models against a simple baseline using appropriate evaluation metrics and a time-based train-validation-test strategy.
5. To examine exchange-rate volatility and evaluate the ability of volatility models to estimate future uncertainty.
6. To investigate differences in forecasting performance across currencies, forecasting horizons and relevant market periods.
7. To convert exchange-rate forecasts into estimated tuition costs in Sri Lankan rupees and construct prediction intervals to represent uncertainty in the estimated costs.
8. To interpret the findings from a financial planning perspective and identify the limitations of the forecasting approaches.

1.5 Significance of the Study

This study is significant because it connects financial time-series forecasting with a practical financial problem faced by Sri Lankan students and their families. By examining exchange rates in the context of overseas tuition payments, the research demonstrates how forecasting techniques can be applied to a real-world financial planning challenge.

From a practical perspective, exchange-rate forecasts can provide an indication of the possible LKR cost of a future tuition payment. Estimates for five and twenty reporting days ahead may help families understand how short-term exchange-rate movements could affect an upcoming

payment. Prediction intervals can further illustrate the range of possible costs and the uncertainty surrounding a point estimate.

From an analytical perspective, the study compares traditional statistical forecasting techniques with machine-learning methods using a consistent evaluation framework. This comparison can provide insight into whether more complex models offer meaningful improvements over a simple baseline and whether performance varies across currencies and forecasting horizons. The analysis of stationarity, autocorrelation and volatility also helps explain the characteristics of financial time series that influence model selection and forecast performance.

The study may also contribute to the practical understanding of forecast uncertainty by translating statistical results into monetary terms. Rather than presenting exchange-rate forecasts only as numerical predictions, the tuition-cost application illustrates how forecast errors and uncertainty can affect financial decisions.

However, the forecasts are intended to support planning rather than guarantee future exchange rates or replace professional financial advice. Their usefulness depends on the quality of the data, the reliability of the models and the extent to which historical patterns remain relevant in the future.

1.6 Scope of the Study

The study focuses on forecasting selected exchange rates relevant to overseas tuition payments by Sri Lankan students. The currencies considered are USD/LKR, GBP/LKR and EUR/LKR, representing the number of Sri Lankan rupees required to purchase one unit of the corresponding foreign currency.

The study period is 3 January 2000 to 31 December 2025. Historical exchange-rate observations are analysed to identify trends, statistical characteristics and volatility patterns. The forecasting analysis considers two horizons: five and twenty reporting days ahead.

The modelling framework includes a baseline forecasting method, traditional time-series forecasting, machine-learning approaches and a separate volatility-modelling component. Model performance is assessed using a chronological training, validation and testing framework to preserve the temporal order of observations and reduce the risk of information leakage from future data.

The study also includes an application in which selected foreign-currency tuition amounts are converted into estimated LKR costs using the forecast exchange rates. Prediction intervals are used to communicate the uncertainty associated with these estimates.The scope is limited to exchange-rate forecasting and its application to illustrative tuition payments. The study does not attempt to forecast university tuition-fee changes, living expenses, accommodation costs, international transfer charges or other components of the total cost of studying overseas. Unless explicitly included in the modelling framework, macroeconomic and political factors are not treated as separate forecasting inputs.

Chapter 2: Data and Preprocessing

2.1 Data source and overview

This study uses daily exchange-rate data from the Federal Reserve Economic Data (FRED) database. The aim of the data preparation stage is to produce a clean, consistent dataset of the US dollar (USD), British pound sterling (GBP) and euro (EUR) expressed in Sri Lankan rupees (LKR). Three FRED series were used:

Table 2.1: *FRED series used in the study*

FRED series	Description	Unit
DEXSLUS	Sri Lankan rupees per US dollar	LKR/USD
DEXUSUK	US dollars per British pound	USD/GBP
DEXUSEU	US dollars per euro	USD/EUR

Source: Federal Reserve Bank of St. Louis, FRED.

2.2 Loading and inspecting the raw data

The three raw files were loaded from the data/raw/ folder and inspected for their dimensions, column names and first observations. The files differ in length and coverage: the euro series has 7,245 rows beginning in January 1999, the pound series has 14,550 rows beginning in January 1971, and the rupee series has 14,029 rows beginning in January 1973. Each file contains an observation date and a single rate column. The rupee file uses a month/day/year date format, while the other two use year-month-day. This inspection identified the formatting differences that had to be handled before the series could be combined.

```
File: Exchange rate Euro.csv
Rows and columns: 7245 2
Column names: observation_date, DEXUSEU
  observation_date DEXUSEU
1       1999-01-04  1.1812
2       1999-01-05  1.1760
3       1999-01-06  1.1636
4       1999-01-07  1.1672
5       1999-01-08  1.1554
6       1999-01-11  1.1534

File: Exchange rate pound.csv
Rows and columns: 14550 2
Column names: observation_date, DEXUSUK
  observation_date DEXUSUK
1       1971-01-04  2.3938
2       1971-01-05  2.3949
3       1971-01-06  2.3967
4       1971-01-07  2.3963
5       1971-01-08  2.3972
6       1971-01-11  2.3992
```

```
File: Exchange rate_usd.csv
Rows and columns: 14029 2
Column names: observation_date, DEXSLUS
  observation_date DEXSLUS
1         1/2/1973  6.7114
2         1/3/1973  6.7114
3         1/4/1973  6.6890
4         1/5/1973  6.7114
5         1/8/1973  6.7114
6         1/9/1973  6.6890

> # Show the exact keys available
> for (nm in names(raw_tables)) print(nm)
[1] "Exchange rate Euro.csv"
[1] "Exchange rate pound.csv"
[1] "Exchange rate_usd.csv"
```

2.3 Standardising formats and restricting the study period

Because the three files use different date formats, each file's date column was converted to a standard date type using its own format, and each rate column was converted to numeric. Any invalid date or non-numeric rate was set to missing so that it could be counted in the quality checks. The series were then restricted to the study period, 3 January 2000 to 31 December 2025, sorted chronologically and reduced to the date and rate columns. The following checks were recorded for each series: number of rows retained, first and last dates, invalid dates, duplicate dates and missing rates.

After filtering, each series contained 6,783 rows. There were no invalid or duplicate dates in any series, and each series contained 265 missing rates.

Table 2.2: *Data-quality checks after restricting to 3 January 2000 – 31 December 2025*

Series	Rows	Invalid dates	Duplicate dates	Missing rates
USD (DEXSLUS)	6,783	0	0	265
GBP (DEXUSUK)	6,783	0	0	265
EUR (DEXUSEU)	6,783	0	0	265

2.4 Merging the series and checking missing dates

The three cleaned series were merged on the observation date using an outer join, which keeps every date that appears in any series. This makes the pattern of missing data visible instead of silently dropping dates during the merge. The merged data were then checked to separate dates where all three rates are missing from dates where only some are missing. There were 265 dates on which all three rates were missing, and no dates on which only some were missing. The missing dates correspond to US public holidays, such as 17 January 2000 and 21 February 2000, on which the source reports no rate. Because the three series are missing on exactly the same dates, no series needed separate treatment.

```
> cat("All three missing:", sum(n_missing == 3), "\n")
All three missing: 265

> cat("Only some missing:", sum(n_missing > 0 & n_missing < 3), "\n")
Only some missing: 0

> # Preview dates containing missing rates
> print(head(combined[n_missing > 0, ], 10))
          date DEXSLUS DEXUSUK DEXUSEU
11  2000-01-17      NA      NA      NA
36  2000-02-21      NA      NA      NA
106 2000-05-29      NA      NA      NA
132 2000-07-04      NA      NA      NA
176 2000-09-04      NA      NA      NA
201 2000-10-09      NA      NA      NA
234 2000-11-23      NA      NA      NA
256 2000-12-25      NA      NA      NA
261 2001-01-01      NA      NA      NA
271 2001-01-15      NA      NA      NA
```

2.5 Missing-value treatment and exchange-rate conversion

Handling missing observations. Dates with missing rates were excluded, giving a complete, date-aligned dataset. No exchange rates were estimated or imputed. This complete-case approach keeps the observations consistent across currencies. It does leave gaps between reporting days, mostly weekends and holidays, and these gaps are examined in the exploratory analysis.

Converting rates to rupees. The source series do not all express rates in rupees: DEXUSUK and DEXUSEU give US dollars per pound and per euro. GBP/LKR and EUR/LKR were therefore calculated by cross-rate multiplication:

- USD/LKR = DEXSLUS
- GBP/LKR = DEXSLUS × DEXUSUK
- EUR/LKR = DEXSLUS × DEXUSEU

The results are rupees per unit of foreign currency for all three currencies. These are derived exchange rates and should not be interpreted as the selling rates quoted by banks.

Validation. The final dataset was checked to confirm that all rates are positive, that there are no missing values or duplicate dates, and that the columns are date, USD, GBP and EUR. The result contains 6,518 reporting days. For example, on 3 January 2000 the rates were USD 72.30, GBP 117.63 and EUR 73.42 LKR.

```
> cat("Reporting days:", nrow(rates), "\n")
Reporting days: 6518

> print(head(rates))
        date   USD      GBP      EUR
1 2000-01-03 72.30 117.6321 73.42065
2 2000-01-04 72.65 118.9281 74.89489
3 2000-01-05 72.95 119.7474 75.39383
4 2000-01-06 72.95 120.1851 75.31358
5 2000-01-07 73.15 119.8490 75.30061
6 2000-01-10 73.30 120.0214 75.14716

> cat("\nMissing values:\n")

Missing values:

> print(colSums(is.na(rates)))
date  USD  GBP  EUR
   0    0    0    0

> cat("\nDuplicate dates:", sum(duplicated(rates$date)), "\n")

Duplicate dates: 0

> cat("\nData types:\n")

Data types:

> print(sapply(rates, function(x) class(x)[1]))
     date       USD       GBP       EUR
   "Date" "numeric" "numeric" "numeric"

> stopifnot(!anyNA(rates[, c("USD", "GBP", "EUR")]))

> stopifnot(all(rates[, c("USD", "GBP", "EUR")] > 0))
```

2.6 Exporting the processed dataset

The cleaned dataset was saved as data/processed/exchange_rates_lkr.csv without row numbers, since the date is stored as a column. A final check confirmed the number of observations (6,518) and the period covered (3 January 2000 to 31 December 2025). This file is the input to all later stages of the project.

```
> # Confirm the sample size and date range
> cat("Rows:", nrow(rates), "\n")
Rows: 6518

> cat("Period:", format(min(rates$date)), "to", format(max(rates$date)), "\n")
Period: 2000-01-03 to 2025-12-31

> print(head(rates))
        date   USD      GBP      EUR
1 2000-01-03 72.30 117.6321 73.42065
2 2000-01-04 72.65 118.9281 74.89489
3 2000-01-05 72.95 119.7474 75.39383
4 2000-01-06 72.95 120.1851 75.31358
5 2000-01-07 73.15 119.8490 75.30061
6 2000-01-10 73.30 120.0214 75.14716
```

2.7 Summary

The preparation process loads three FRED series, standardizes their date and numeric formats, restricts them to 2000–2025, aligns them by date, removes the 265 dates on which no rates were reported, and converts all rates to rupees per unit of foreign currency. The result is a complete, validated dataset of 6,518 daily observations for USD/LKR, GBP/LKR and EUR/LKR.

Chapter 3: Exploratory Data Analysis (EDA)

The exploratory analysis examines the processed dataset of USD/LKR, GBP/LKR and EUR/LKR rates to understand its structure, trends, variability and statistical properties before any forecasting model is built. Because the GBP and EUR series are derived cross-rates and not bank selling rates, they are treated as indicators of the exchange rate rather than prices that could actually be transacted.

3.1 Dataset structure and data quality

The processed dataset contains 6,518 daily observations from 3 January 2000 to 31 December 2025, with columns for date, USD, GBP and EUR. The data types are a date column and three numeric columns. There are no missing values, no duplicate dates and no zero or negative rates, and the number of unique dates equals the number of rows. These checks confirm that the data used in the analysis is the dataset produced in the preparation stage.

```
> cat("Dataset shape:", nrow(df), "x", ncol(df), "\n")
Dataset shape: 6518 x 4

> cat("Study period:", format(min(df$date)), "to", format(max(df$date)), "\n")
Study period: 2000-01-03 to 2025-12-31

> print(head(df))
        date   USD      GBP      EUR
1 2000-01-03 72.30 117.6321 73.42065
2 2000-01-04 72.65 118.9280 74.89489
3 2000-01-05 72.95 119.7474 75.39383
4 2000-01-06 72.95 120.1851 75.31358
5 2000-01-07 73.15 119.8490 75.30061
6 2000-01-10 73.30 120.0214 75.14716
```

```
> # ---- 3. Dataset Structure and Data Quality ----
>
> cat("Data types:\n")
Data types:

> print(data.frame(dtype = sapply(df, function(x) class(x)[1])))
       dtype
date    Date
USD  numeric
GBP  numeric
EUR  numeric

> quality_summary <- data.frame(missing_values = colSums(is.na(df[required_cols])))

> cat("\nMissing values:\n")

Missing values:

> print(quality_summary)
     missing_values
date              0
USD               0
GBP               0
EUR               0
```

```
Duplicate dates: 0

> cat("Non-positive exchange-rate values:\n")
Non-positive exchange-rate values:

> print(data.frame(count = colSums(df[currency_cols] <= 0)))
    count
USD     0
GBP     0
EUR     0

> cat("\nRows:", nrow(df), "\n")

Rows: 6518

> cat("Unique dates:", length(unique(df$date)), "\n")
Unique dates: 6518

> cat("Date range:", format(min(df$date)), "to", format(max(df$date)), "\n")
Date range: 2000-01-03 to 2025-12-31

> stopifnot("Missing dates found." = !anyNA(df$date))

> stopifnot("Duplicate dates found." = !anyDuplicated(df$date))

> stopifnot("Missing exchange rates found." = !anyNA(df[currency_cols]))

> stopifnot("Non-positive exchange rates found." = all(df[currency_cols] > 0))
```

3.2 Descriptive statistics of exchange-rate levels

The rupee depreciated against all three currencies over the period. USD/LKR averaged 154.35 (standard deviation 74.73), with a minimum of 72.30 and a maximum of 365.43. GBP/LKR averaged 223.67 (range 109.89 to 454.09) and EUR/LKR averaged 180.36 (range 65.50 to 398.15). The ranges are large relative to the means, which reflects the long-term upward trend and the sharp rise in 2022. Because the series are in different units of rupees, they are compared by their behaviour rather than by raw size.

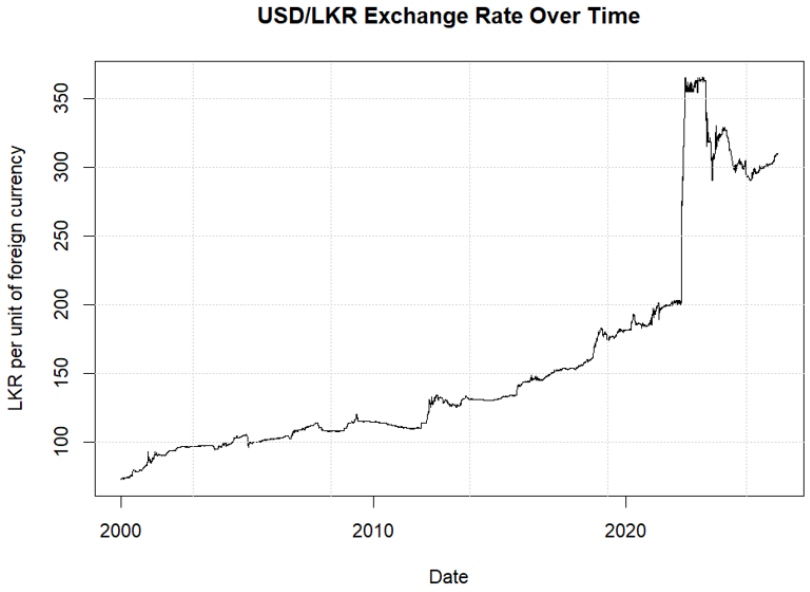
```
> # ---- 4. Descriptive Statistics of Exchange-Rate Levels ----
>
> level_summary <- describe_cols(df[currency_cols])

> level_summary <- cbind(level_summary,
+                        range = level_summary[, "max"] - level_summary[, "min"])

> print(round(level_summary, 4))
    count     mean     std      min      25%      50%      75%      max    range
USD  6518 154.3487 74.7286  72.3000 103.9700 129.5000 179.2375 365.4300 293.1300
GBP  6518 223.6693 81.2506 109.8888 178.2313 204.4382 229.6925 454.0928 344.2040
EUR  6518 180.3609 78.0686  65.4984 132.5818 161.9880 199.4179 398.1467 332.6483
```

3.3 Trends over time

The line charts show a steady long-term rise in all three rates. USD/LKR rose from about 72 in January 2000 to about 309 at the end of 2025. The sharpest increase occurred in 2022, when USD/LKR rose from about 203 at the end of 2021 to about 363 at the end of 2022. The shape of the series, trending upward with one abrupt jump, suggests the data are not stationary and contain a structural break, which is tested in the later sections.



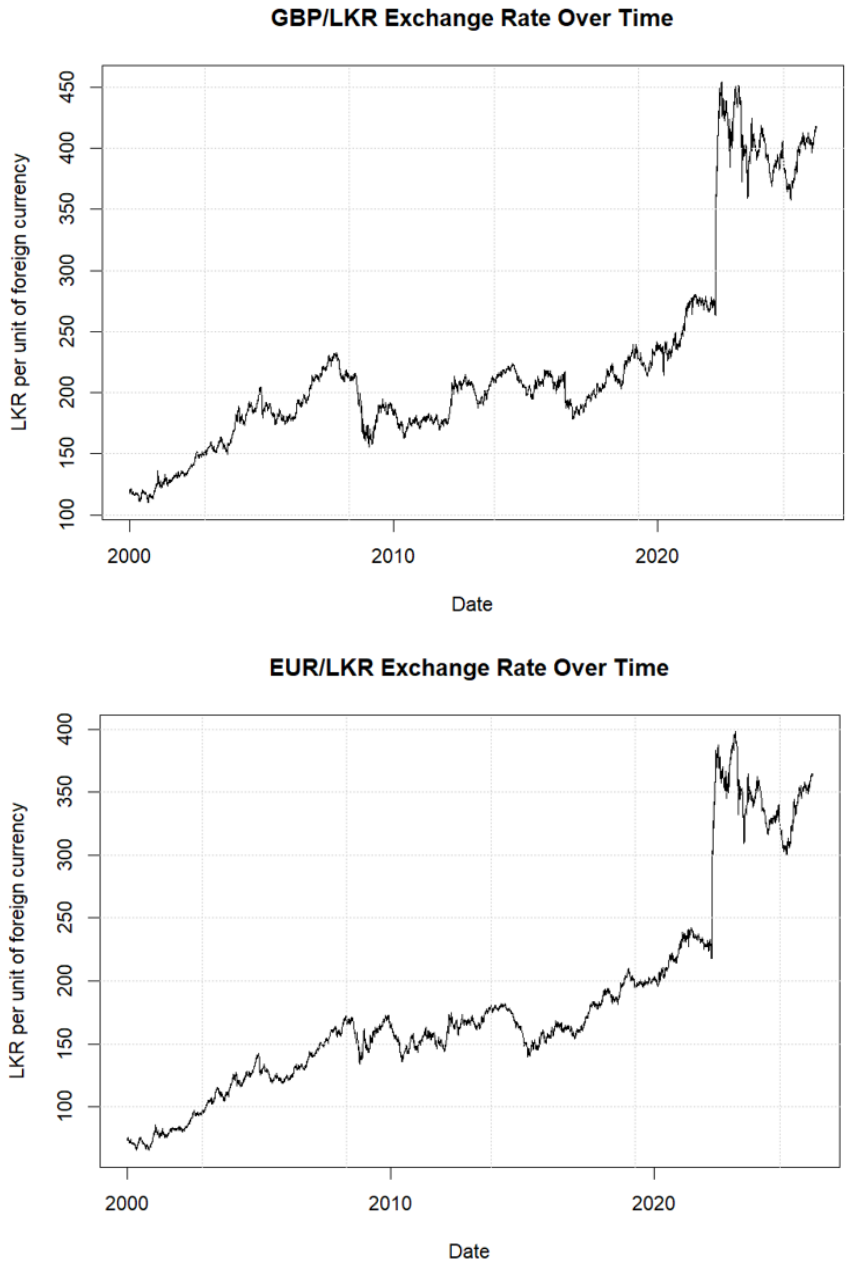


Figure 3.1: Daily exchange rates in LKR per unit of foreign currency, 2000–2025.

3.4 Daily returns

Exchange-rate levels trend, so the analysis also examines daily returns. First, daily percentage changes were calculated as 100 × (P_t / P_{t-1} − 1). Later diagnostics use log returns, 100 × (ln P_t − ln P_{t−1}), which are additive over time, approximately equal to percentage changes for small moves, and the standard input for ARIMA and GARCH models. Both are calculated between consecutive available observations, so they may span a weekend or holiday.

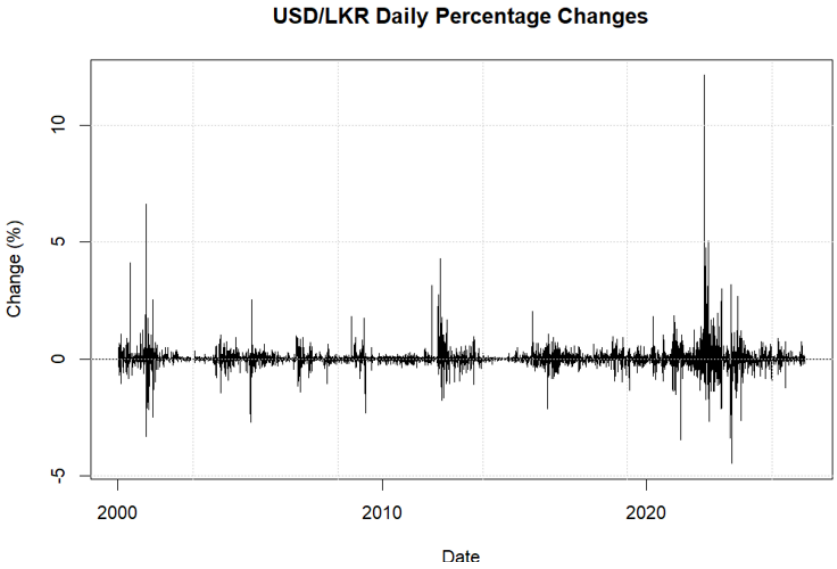
Average daily changes are small and positive (0.023% for USD, 0.022% for GBP and 0.027% for EUR), consistent with gradual depreciation. Variability differs by currency: the standard

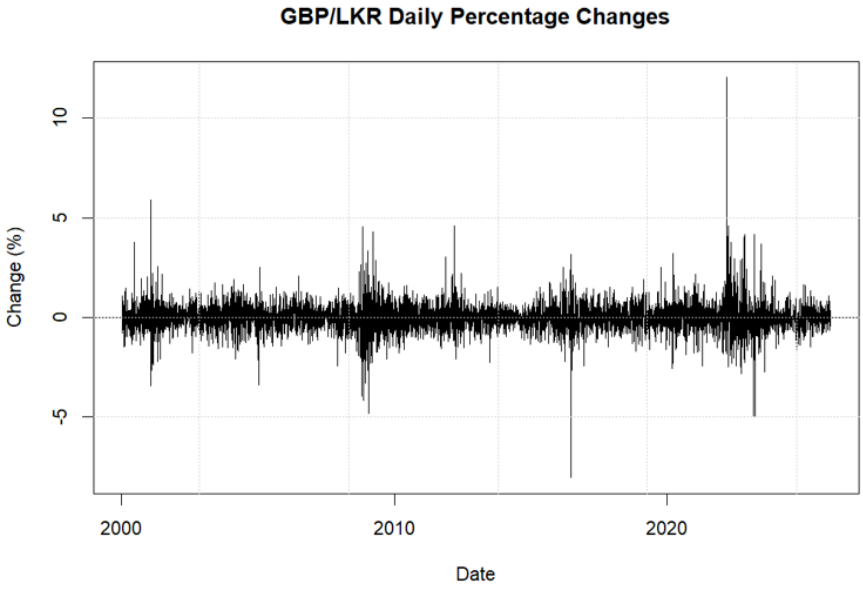
deviation is 0.42% for USD against 0.72% for GBP and 0.71% for EUR. The largest daily rise is over 12% for all three currencies, while the largest falls are 4.5% (USD), 8.0% (GBP) and 4.5% (EUR).

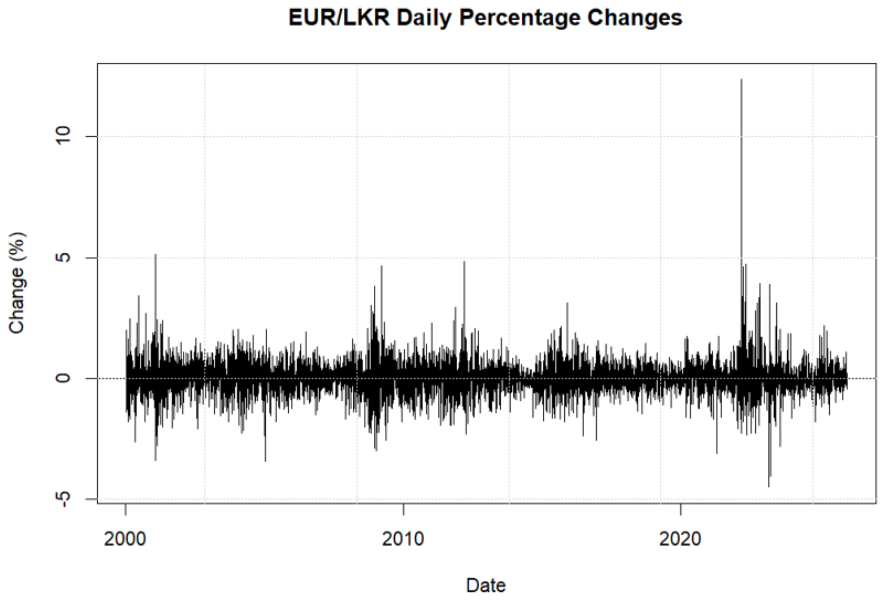
Table 3.1: *Summary of daily log returns (%) and extreme daily changes*

Statistic	USD	GBP	EUR
Mean daily change (%)	0.023	0.022	0.027
Standard deviation (%)	0.42	0.72	0.71
Skewness (log returns)	5.76	0.62	1.03
Kurtosis (log returns)	148.9	21.8	19.2
Largest daily rise	> 12%	> 12%	> 12%
Largest daily fall	4.5%	8.0%	4.5%

Jarque–Bera test rejects normality for all three currencies ($p \approx 0$).

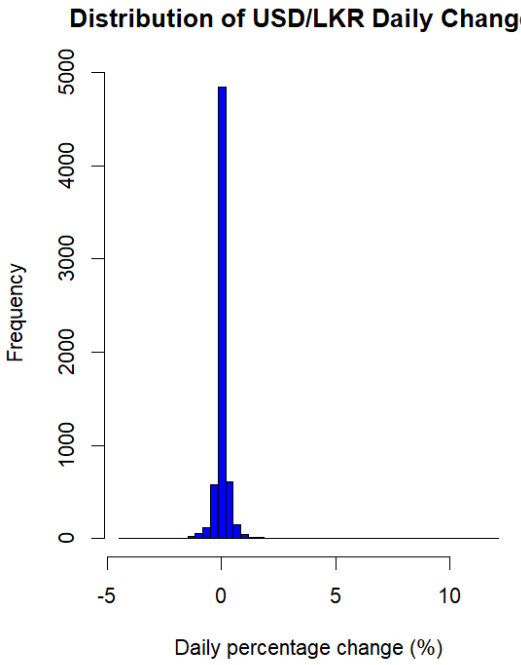


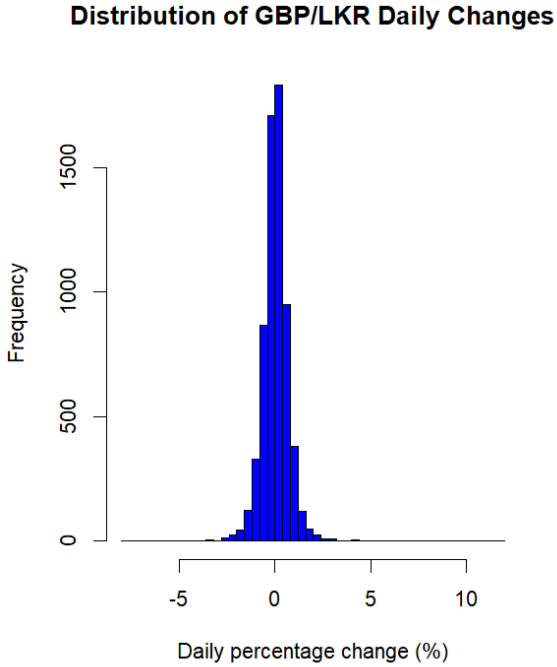


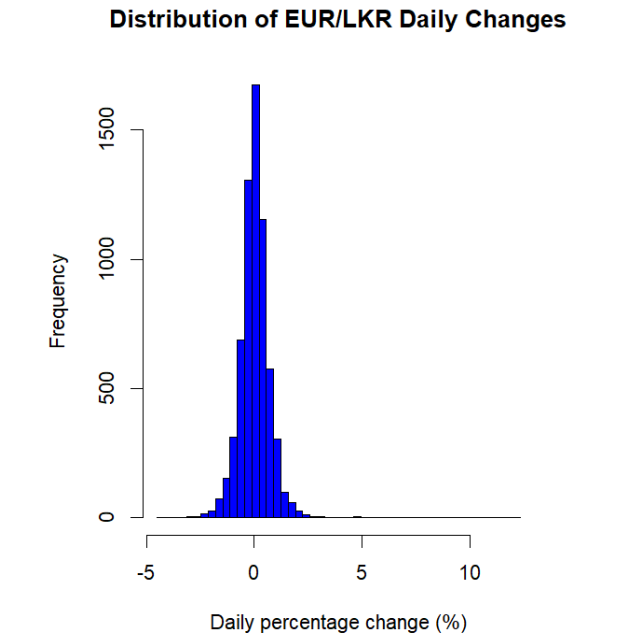


3.5 Distribution and normality of returns

The histograms show that daily changes cluster tightly around zero with a few very large values. This is confirmed by the Jarque–Bera test, which rejects normality for all three currencies ($p \approx 0$). Skewness of log returns is 5.76 for USD, 0.62 for GBP and 1.03 for EUR, and kurtosis is 148.9, 21.8 and 19.2 respectively, far above the value of 3 for a normal distribution. The distribution therefore has very heavy tails: extreme daily moves occur much more often than a normal model would imply. The QQ plots should show the same pattern, with points departing from the reference line at both ends. This has a direct consequence for the forecasting stage, because prediction intervals based on a normal distribution would be too narrow.








```
> print(vol_norm, row.names = FALSE)
 series ARCH-LM p JB p   skew kurtosis
    USD         0    0 5.7608 148.8821
    GBP         0    0 0.6185  21.7917
    EUR         0    0 1.0306  19.2145
```

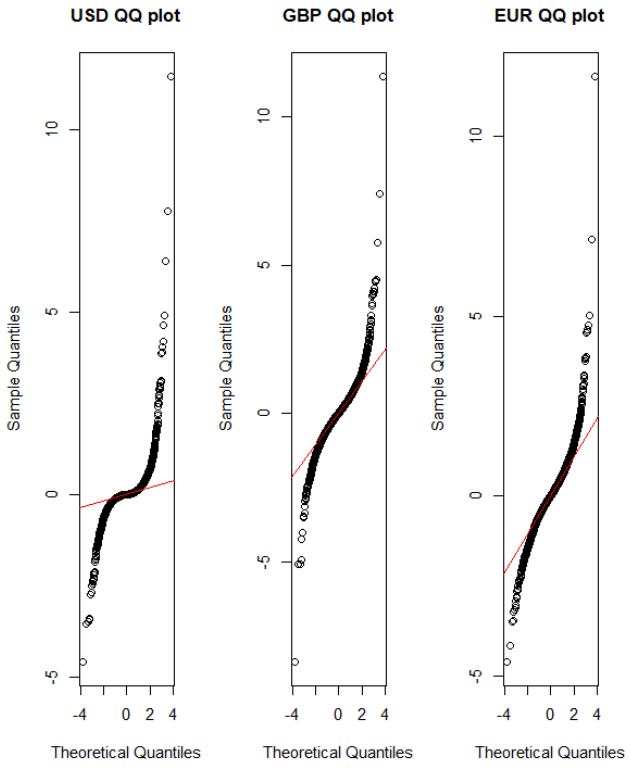
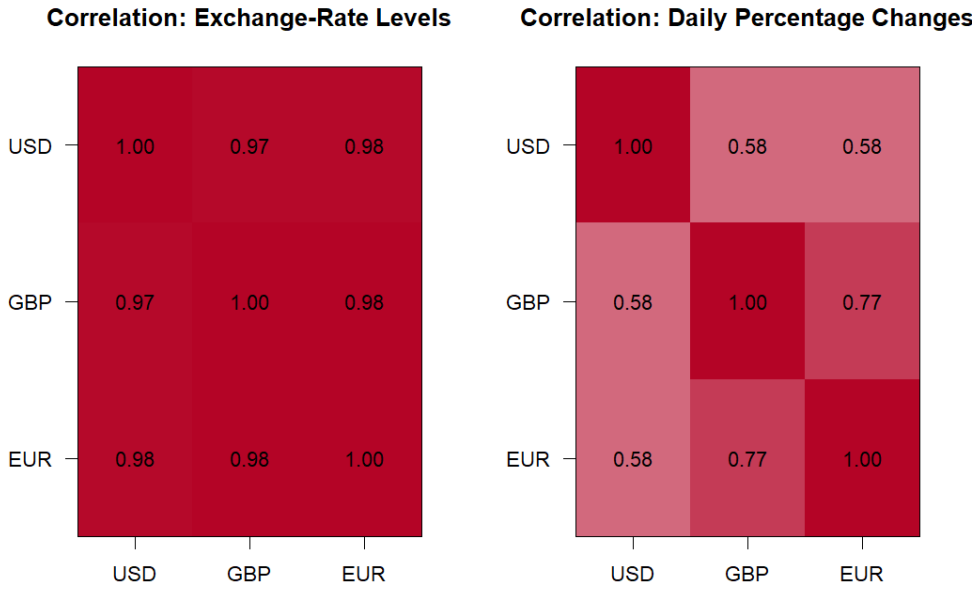


Figure 3.3: Distribution of daily returns and normal QQ plots.

3.6 Correlation between currencies

The levels of the three series are highly correlated (0.97–0.98), as expected for trending series that share the USD/LKR component. This high correlation should not be treated as independent evidence of a relationship. Correlations of daily returns are lower: 0.58 between USD and each of GBP and EUR, and 0.77 between GBP and EUR. GBP and EUR move together more closely because both are driven by their exchange rates against the US dollar, while the USD/LKR series is largely driven by local rupee conditions.



3.7 Rolling volatility and volatility clustering

A 20-observation rolling standard deviation shows that volatility is concentrated in particular periods instead of being constant. Rolling volatility peaked in March 2022, at roughly 3% for all three currencies, and 2022 had the highest average volatility. The ARCH-LM test rejects the null hypothesis of no ARCH effects for all three currencies ($p \approx 0$), which confirms volatility clustering: large movements tend to be followed by large movements. This supports using GARCH-type models for volatility.

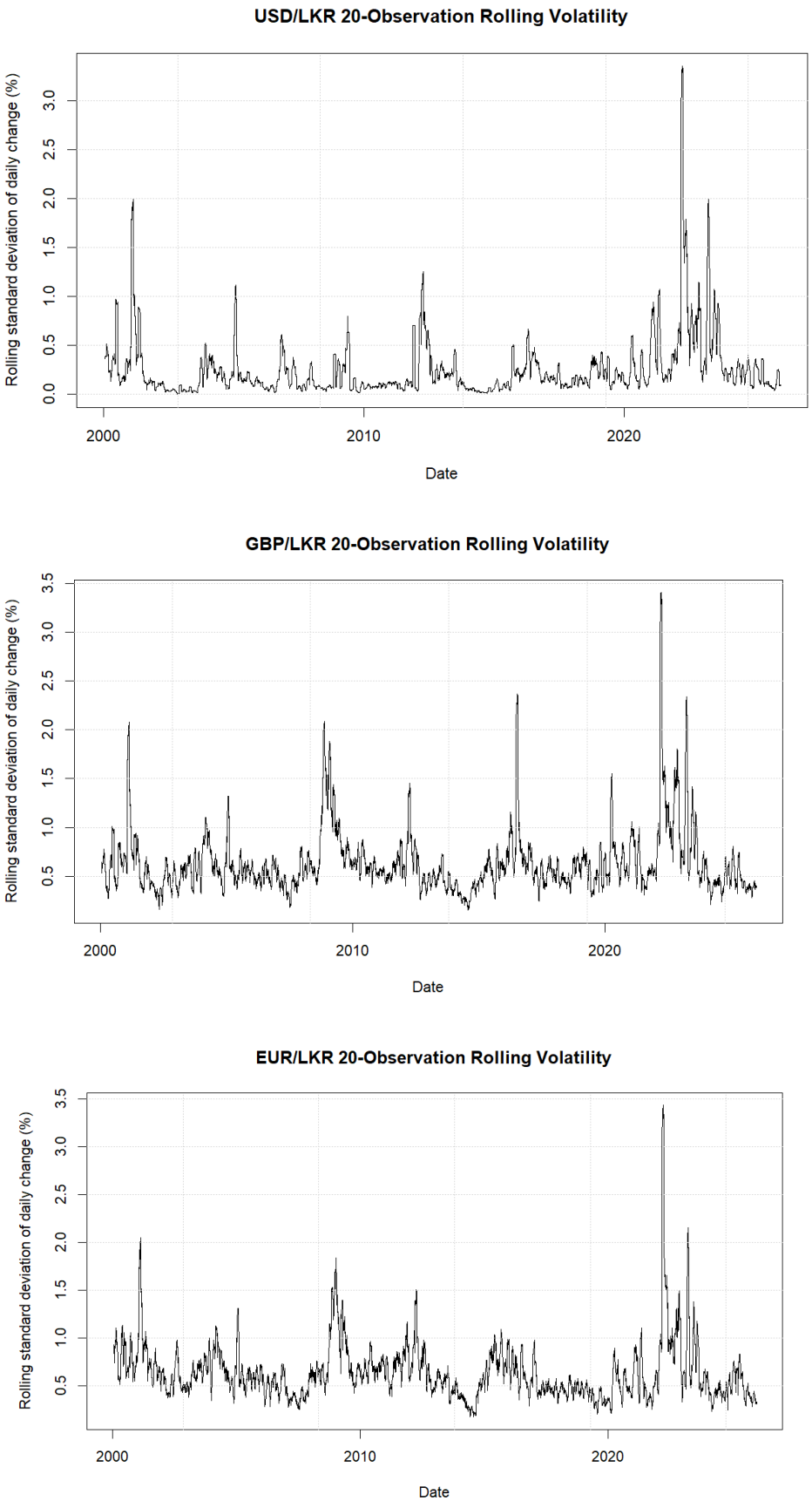


Figure 3.5: 20-observation rolling standard deviation of daily returns.

3.8 Stationarity

Stationarity was tested with the Augmented Dickey–Fuller (ADF) test, whose null hypothesis is a unit root, and the KPSS test, whose null hypothesis is stationarity. Using both gives stronger evidence, because the two tests have opposite null hypotheses.

For exchange-rate levels, the ADF test does not reject the unit root (p-values of 0.88 for USD, 0.88 for GBP and 0.91 for EUR on levels, and 0.90, 0.79 and 0.75 for log levels), and KPSS rejects stationarity ($p \leq 0.01$). The levels are therefore non-stationary. For daily changes, the ADF statistics are strongly negative (−10.27 for USD, −13.52 for GBP and −14.96 for EUR; $p \approx 0$) and KPSS does not reject stationarity ($p \geq 0.10$), so returns are stationary. This is why the forecasting models are fitted to returns instead of levels. The ADF test can be weakened by structural breaks, so the 2022 shock should be kept in mind when interpreting these results. KPSS p-values are bounded between 0.01 and 0.10 by the test's critical-value table.

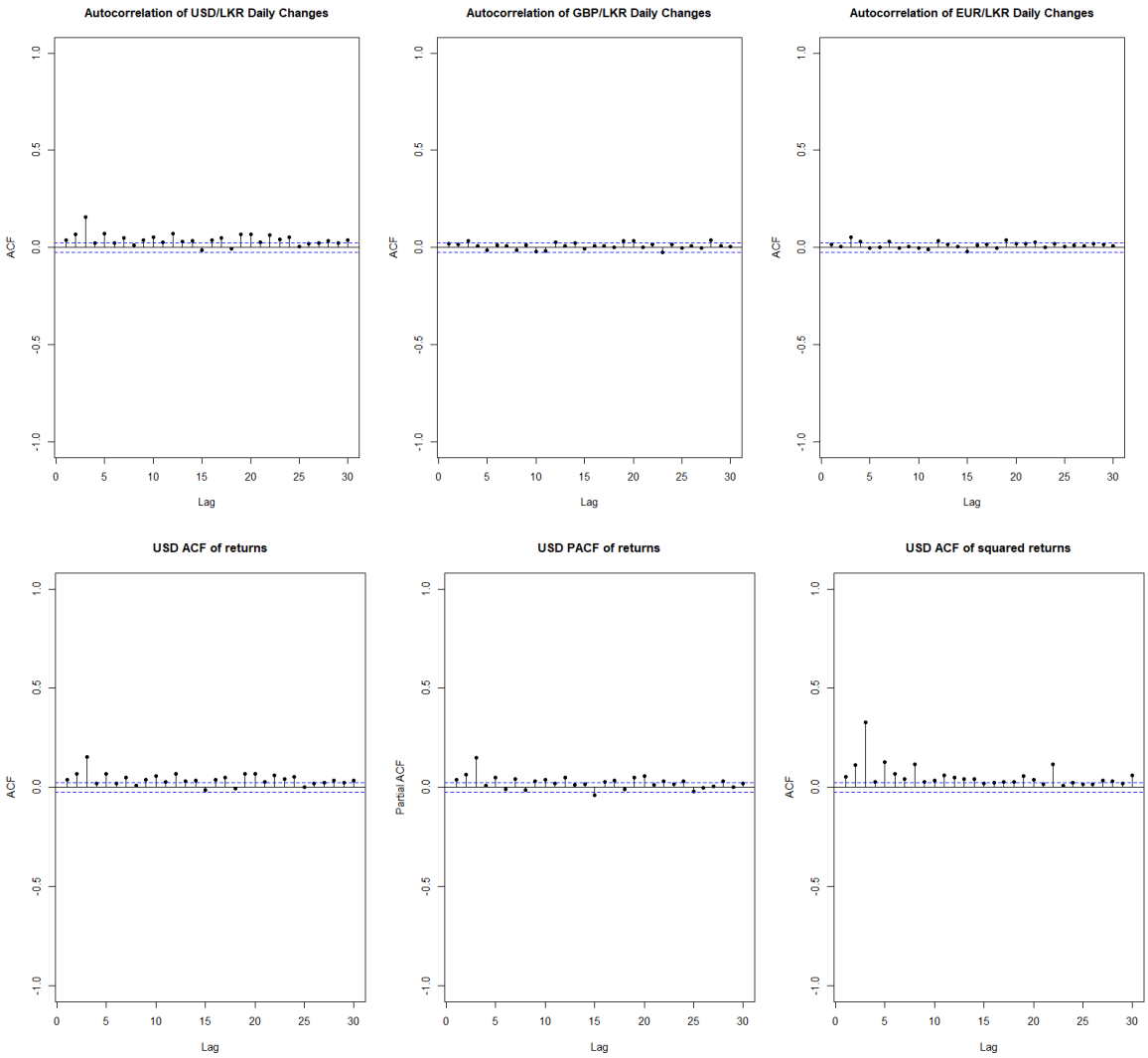
```
> print(adf_results, row.names = FALSE)
              series ADF statistic p-value used lags observations 5% critical value
            USD_level       -0.5379  0.8844        32         6485            -2.862
            GBP_level       -0.5711  0.8774        32         6485            -2.862
            EUR_level       -0.3712  0.9148        34         6483            -2.862
 USD_daily_change_pct      -10.2662  0.0000        35         6481            -2.862
 GBP_daily_change_pct      -13.5182  0.0000        27         6489            -2.862
 EUR_daily_change_pct      -14.9601  0.0000        21         6495            -2.862
```

```
> print(stationarity, row.names = FALSE)
 series  transform  ADF p KPSS p
    USD  log level 0.8953   0.01
    USD log return 0.0000   0.10
    GBP  log level 0.7896   0.01
    GBP log return 0.0000   0.10
    EUR  log level 0.7455   0.01
    EUR log return 0.0000   0.10
```

3.9 Autocorrelation

The autocorrelation plots show little linear structure in returns. For GBP and EUR, almost no autocorrelation or partial autocorrelation is significant, so past returns carry little information about future returns. USD/LKR shows small positive autocorrelation at several lags, with the clearest spike at lag 3 (about 0.15).

In contrast, the autocorrelation of squared returns is large and persistent for all three currencies, with a prominent spike at lag 3 (0.33 for USD, 0.26 for GBP and 0.27 for EUR). This is the pattern expected under volatility clustering. To check whether this lag-3 spike depends on the 2022 shock, the calculation was repeated excluding 2022. The squared-return autocorrelation falls to 0.09 for USD, 0.10 for GBP and 0.12 for EUR, and the USD lag-3 return autocorrelation falls from 0.15 to 0.04. The spike is therefore largely driven by the March 2022 shocks, although volatility clustering remains in the rest of the data. Excluding a year removes a block from the middle of the series, so this is a rough robustness check rather than a formal test.


Autocorrelation of USD/LKR Daily Changes
Autocorrelation of GBP/LKR Daily Changes
Autocorrelation of EUR/LKR Daily Changes
USD ACF of returns
USD PACF of returns
USD ACF of squared returns
ACF
Partial ACF
Lag

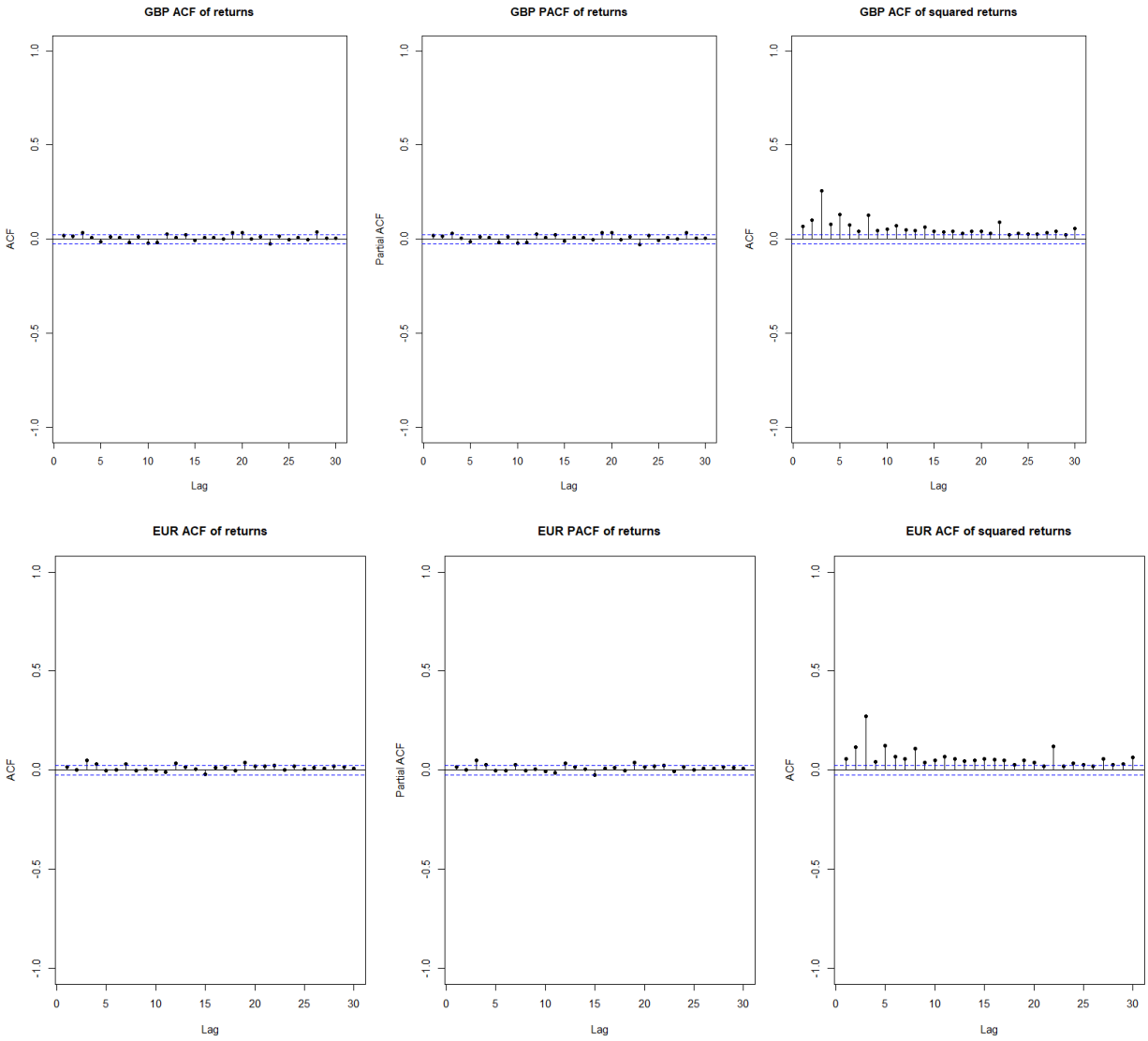


Figure 3.6: Autocorrelation of daily changes, returns and squared returns.

3.10 Reporting-day gaps and forecast horizons

The data are recorded on reporting days, not every calendar day. Of the 6,517 gaps between consecutive observations, 5,088 are one day, 1,168 are three days (weekends), 184 are four days, 73 are two days and 4 are five days. The mean gap is 1.46 days and the maximum is five. A horizon of five reporting days therefore corresponds to roughly one calendar week, and 20 reporting days to roughly one month. Forecast horizons in this project are defined in reporting days.

Table 3.4: *Gaps between consecutive reporting days*

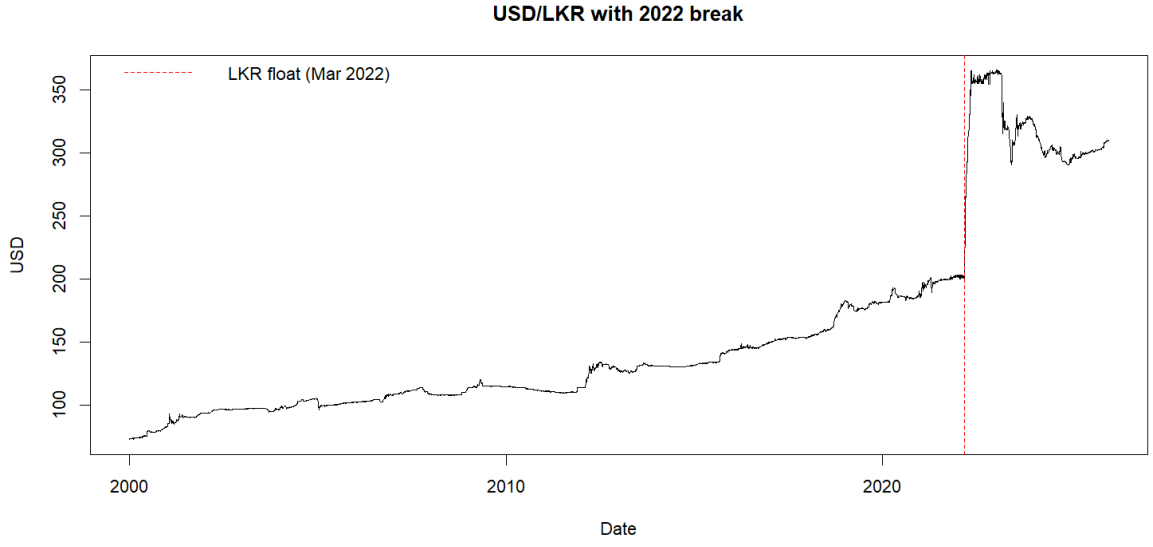
Gap (calendar days)	Count	Typical cause
1	5,088	Consecutive weekdays
2	73	Single missing weekday
3	1,168	Weekends
4	184	Holidays / long weekends
5	4	Holidays

Total 6,517 gaps; mean gap 1.46 days; maximum 5 days.

3.11 Structural break: the 2022 rupee float

In March 2022 the rupee was allowed to float after a currency crisis, and it depreciated sharply. The level plots with the float date marked show the jump clearly, and the return statistics differ markedly across three periods. For USD/LKR, the daily standard deviation was 0.30% in 2000–2021, 1.30% in 2022 and 0.47% in 2023–2025, so volatility fell after the float but remained higher than before. The largest daily rise in 2022 was about 11.5% for USD. GBP and EUR volatility was about 0.66% both before and after the float, because it is dominated by movements in GBP/USD and EUR/USD. Average returns were positive before 2023 (depreciation) and slightly negative in 2023–2025 (appreciation). This break suggests that a model trained mainly on earlier data may struggle to predict later behaviour, which motivates comparing model performance before and after the break.

```
    2000-2021 | mean 2000-2021 | std 2000-2021 | min 2000-2021 | max 2022 (crisis) | mean
USD             0.019           0.300           -3.539          6.410                0.233
GBP             0.015           0.666           -8.374          5.735                0.189
EUR             0.021           0.663           -3.498          5.005                0.211
    2022 (crisis) | std 2022 (crisis) | min 2022 (crisis) | max 2023-2025 | mean
USD               1.301               -2.688              11.473          -0.021
GBP               1.506               -2.844              11.366          -0.007
EUR               1.432               -2.361              11.666          -0.009
    2023-2025 | std 2023-2025 | min 2023-2025 | max
USD           0.467           -4.581          3.125
GBP           0.686           -5.088          4.092
EUR           0.663           -4.602          3.817
```



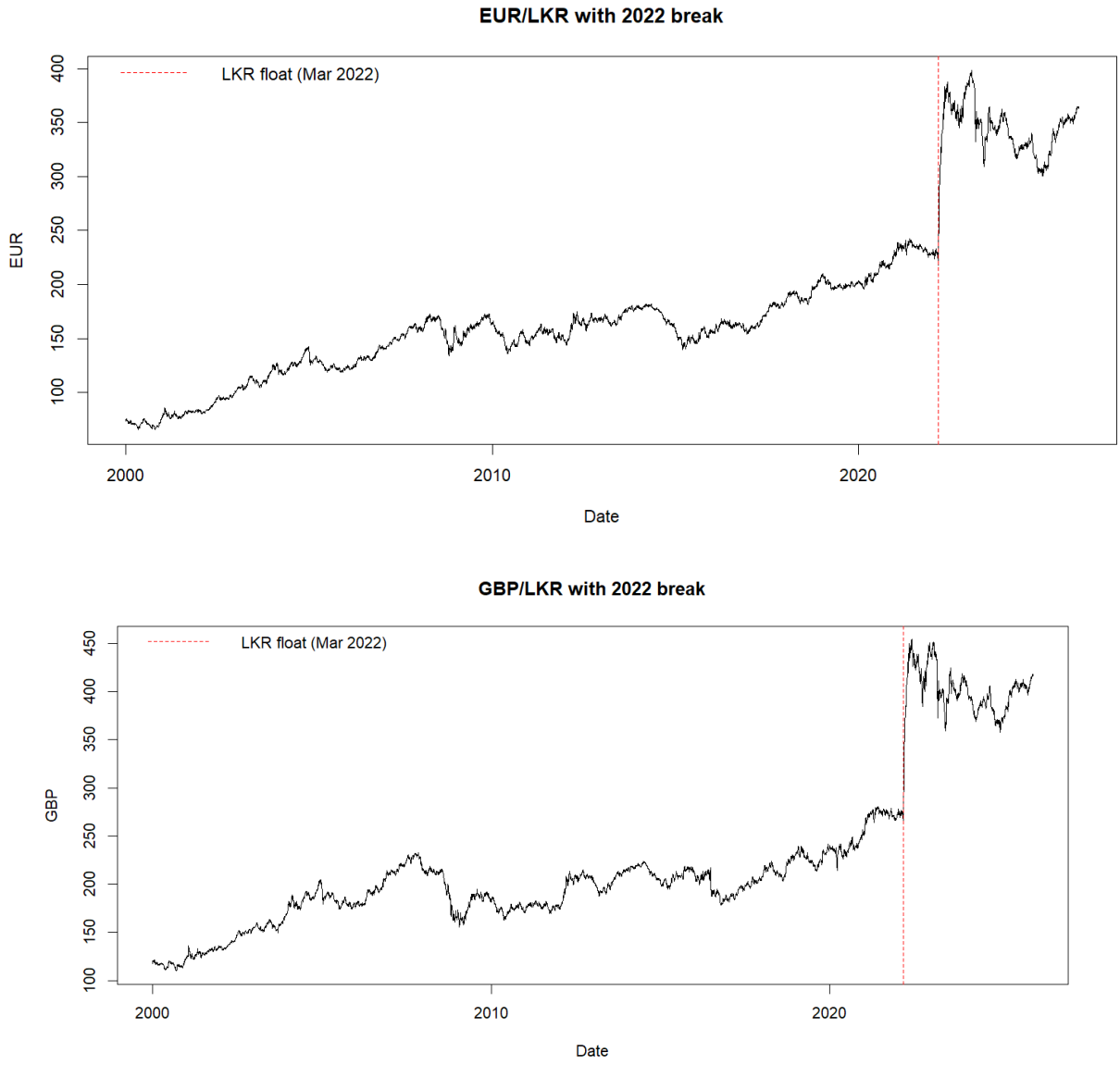


Figure 3.6: Exchange rates with the March 2022 float marked.

3.12 Unchanged rates

USD/LKR has many days with no change in every period, between about 5% and 56% of days each year. The share is highest in 2002 (56%) and in 2014–2015 (about 40%), is 28% in 2022, and falls to about 5–12% in 2023–2025. This is consistent with a closely managed rate and with repeated reporting of the same value. GBP/LKR and EUR/LKR have almost no unchanged days, because they incorporate floating USD rates. These flat stretches can affect stationarity tests, volatility estimates, and the patterns learned by machine-learning models.

	USD	GBP	EUR
2000	0.263	0.012	0.008
2001	0.380	0.004	0.000
2002	0.558	0.004	0.000
2003	0.219	0.004	0.000
2004	0.130	0.000	0.000
2005	0.064	0.004	0.000
2006	0.112	0.000	0.000
2007	0.154	0.012	0.016
2008	0.143	0.004	0.000
2009	0.250	0.000	0.004
2010	0.204	0.000	0.000
2011	0.247	0.000	0.000
2012	0.143	0.004	0.000
2013	0.275	0.000	0.000
2014	0.412	0.000	0.004
2015	0.394	0.000	0.004
2016	0.159	0.004	0.000
2017	0.241	0.004	0.000
2018	0.201	0.000	0.004
2019	0.120	0.004	0.004
2020	0.180	0.000	0.004
2021	0.241	0.004	0.004
2022	0.284	0.004	0.000
2023	0.116	0.004	0.000
2024	0.052	0.000	0.004
2025	0.116	0.000	0.004

3.13 Summary and implications for forecasting

The exploratory analysis shows that the exchange-rate levels are non-stationary and trending, while returns are stationary, heavy-tailed and only weakly autocorrelated. Volatility clusters strongly, and a structural break in 2022 divides the sample into different regimes. These findings lead to five modelling decisions:

1. Models are fitted to returns, not levels.
2. A random-walk baseline is essential, since weak autocorrelation suggests it will be hard to beat, especially for GBP and EUR.
3. Volatility clustering supports GARCH-type volatility forecasts.
4. Heavy tails mean prediction intervals should not assume normality.
5. The 2022 break should be treated explicitly, for example by comparing performance before and after it, and evaluation should use a chronological split with walk-forward forecasts.

Chapter 4: Methodology

4.1 Research design

This study used a quantitative forecasting approach to estimate exchange rates relevant to overseas tuition payments by Sri Lankan students. Traditional statistical models and machine-learning models were compared for USD/LKR, GBP/LKR and EUR/LKR at five- and twenty-reporting-day forecast horizons. A separate volatility-modelling component was included to investigate uncertainty in exchange-rate movements. Forecast exchange rates were subsequently translated into illustrative tuition-payment costs in Sri Lankan rupees.

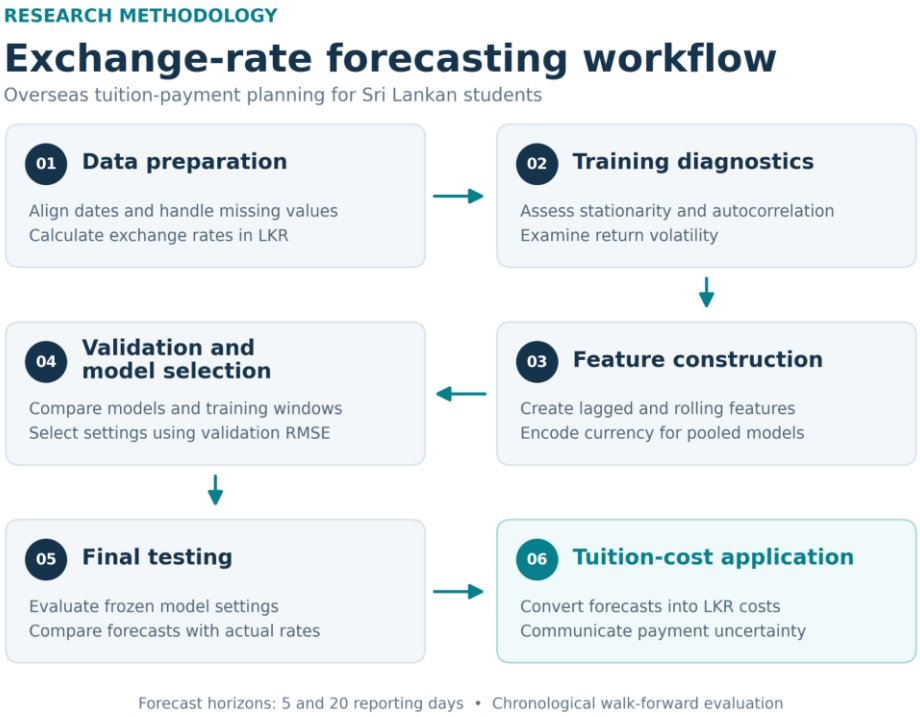


Figure 4.1: *Exchange-rate forecasting workflow.*

4.2 Definition of the forecasting task

The main forecasting target was the number of Sri Lankan rupees required to purchase one unit of USD, GBP or EUR five and twenty reporting days ahead. Reporting days refer to dates with available exchange-rate observations rather than consecutive calendar days. Forecasts were assumed to be produced after the exchange rate at the forecast origin had become available.

The volatility analysis addressed a different target: the variability of exchange-rate returns. This distinction separates forecasts of the future exchange-rate level from forecasts of uncertainty surrounding exchange-rate movements.

Table 4.1: *Components of the forecasting task*

Component	Description
Currencies	USD, GBP and EUR, expressed in LKR
Price target	Exchange rate five or twenty reporting days ahead
Forecast origin	After the current reporting-day rate is available
Forecast target	The exchange rate 5 or 20 reporting days later
Business application	LKR cost of a specified foreign-currency tuition payment

4.3 Log transformation and return calculation

Logarithms were used to represent exchange-rate movements in proportional terms rather than absolute changes in rupees. An increase of one rupee has a different relative meaning when an exchange rate is 75 than when it is 350. Log changes provide a more consistent measure of movement across these different exchange-rate levels.

The transformation also reduces the influence of differences in absolute currency scales, which is useful when combining currencies in pooled machine-learning models. Differences between consecutive log exchange rates were used to calculate log returns. Percentage log returns were used in the diagnostic analysis.

ARIMA models were fitted to log exchange-rate levels with first differencing, while ETS models were fitted to log levels. The supervised models predicted the cumulative log change over each forecast horizon. Their predictions were converted back into exchange rates for evaluation and tuition-cost calculations.

The use of logarithms does not automatically make the series stationary or normally distributed. Stationarity and distributional characteristics were therefore examined separately.

4.4 Chronological data split

The 6,518 reporting-day observations were divided chronologically into three non-overlapping periods. Observations from 2000–2021 formed the initial training set, 2022–2023 formed the validation set used to choose model settings and estimation windows, and 2024–2025 formed the final test set. The test period was not used for any modelling decision.

Table 4.2: *Chronological train–validation–test split*

Period	Years	Reporting days	Purpose
Initial training	2000–2021	5,518	Initial model fitting and training diagnostics
Validation	2022–2023	499	Parameter and estimation-window selection
Final testing	2024–2025	501	Out-of-sample evaluation of frozen models

A chronological split was used instead of random sampling because random splits would allow future observations to inform predictions of the past. The validation period deliberately included the 2022 rupee float (Section 3.11), so that candidate models were compared under a major structural shock as well as the calmer conditions of 2023.

4.5 Walk-forward evaluation

Forecasts were produced by a walk-forward procedure. A forecast origin was moved forward through the validation and test periods in steps of five reporting days. At each origin, models were fitted using only observations available up to that origin and were used to forecast the exchange rate five and twenty reporting days ahead. Each forecast was then compared with the rate actually observed at the target date, and forecasts were assigned to the validation or test period according to their target date.

This design gave 96 validation forecasts and 97 test forecasts for each currency, horizon and model configuration. Because origins were five reporting days apart, twenty-day forecast periods overlap and their errors are dependent; this is considered when interpreting the results (Section 4.15).

4.6 Estimation windows

Two estimation-window approaches were compared for every model family. The **expanding window** used all observations from the start of the sample up to the forecast origin. The **recent window** used only the latest 1,250 reporting days (roughly five years) before the origin. The recent window allows a model to adapt faster after a regime change, while the expanding window uses more information. Both approaches were treated as candidate settings and the better one was chosen on validation RMSE, not on test results.

4.7 Prevention of information leakage

All predictors were constructed using information available at or before the forecast origin. Rolling statistics used trailing windows rather than centered windows. For supervised models, a

training example was included only when its future target had already been observed by the forecast origin.

Missing initial feature values resulting from lag and rolling-window construction were excluded from the relevant model-fitting samples. No full-sample imputation, scaling, or outlier removal was applied. Genuine extreme movements were retained.

4.8 Benchmark and statistical forecasting models

Statistical forecasting models were included to provide interpretable comparisons and establish whether machine-learning complexity produced improvements over simpler alternatives.

4.8.1 Naïve benchmark

The naïve model assumed that the exchange rate at the forecast horizon would equal the latest observed rate. This model served as the primary benchmark because exchange-rate movements can be difficult to predict using past observations alone (Meese and Rogoff, 1983).

4.8.2 ARIMA

ARIMA models were fitted to log exchange-rate levels using first differencing. The candidate specifications were ARIMA(0,1,0), ARIMA(0,1,0) with drift, ARIMA(1,1,0), ARIMA(0,1,1) and ARIMA(1,1,1). Candidate models were examined using AIC, BIC and residual diagnostics. An automatic ARIMA search (Hyndman and Khandakar, 2008) was also used as a cross-check. The final ARIMA configuration and estimation window were selected using validation RMSE rather than information criteria alone.

Table 4.3: *Candidate ARIMA specifications*

Candidate	p	d	q	Drift
ARIMA(0,1,0)	0	1	0	No
ARIMA(0,1,0) with drift	0	1	0	Yes
ARIMA(1,1,0)	1	1	0	No
ARIMA(0,1,1)	0	1	1	No
ARIMA(1,1,1)	1	1	1	No

Numerical AIC/BIC comparisons are reported in Section 5.2.

4.8.3 Exponential smoothing

Exponential-smoothing models were fitted to log exchange-rate levels. Candidates included simple exponential smoothing (SES) and automatically selected non-seasonal ETS specifications. These models provided an alternative way to represent changing levels and trends. Seasonality was not modelled because the training-period seasonal strength was negligible (Section 5.2).

4.8.4 Linear regression

Linear regression was included as a simple supervised-learning comparator. Two feature specifications were evaluated: a return-lag specification and a specification containing the complete constructed feature set (Section 4.9). Both predicted the cumulative log change over the forecast horizon.

4.8.5 Vector autoregression

VAR models were included as a multivariate extension to examine whether joint modelling of the three return series improved forecasts. Lag orders of one and two were evaluated. Forecast returns were accumulated and converted back into exchange-rate levels.

4.9 Machine-learning feature construction

Machine-learning predictors were constructed from historical exchange-rate movements. Return lags represented recent changes, rolling averages represented recent average behaviour, and rolling standard deviations represented recent variability. Additional features measured the difference between the current log exchange rate and its trailing average.

Features were calculated separately for each currency. Return-based inputs reduced dependence on absolute exchange-rate levels.

Table 4.4: *Machine-learning feature set*

Feature group	Specification
Current return	Return observed at the forecast origin
Lagged returns	Lags 1–5, 10 and 20
Rolling return means	5, 20 and 60 observations
Rolling return standard deviations	5, 20 and 60 observations
Log-rate distance from trailing mean	5, 20 and 60 observations

4.10 Machine-learning models

4.10.1 Random Forest

Random Forest regression (Breiman, 2001) was used to capture potentially nonlinear relationships between historical features and future exchange-rate changes. Each model used 300 trees. Minimum node sizes of five and twenty were compared for the separate-currency models. The number of candidate predictors considered at each split was based on the square root of the total number of predictors.

4.10.2 XGBoost

XGBoost regression (Chen and Guestrin, 2016) was used to assess whether sequentially fitted decision trees improved forecasting performance. Maximum tree depths of two and three were compared. The learning rate was fixed at 0.03, with 200 boosting rounds. Row and column sampling ratios were both set to 0.8, and the minimum child weight was set to 10.

Table 4.5: *Parameters of the limited candidate search for Random Forest and XGBoost*

Model	Fixed settings	Candidate values compared
Random Forest	300 trees; mtry = √(number of predictors)	Minimum node size: 5, 20
XGBoost	Learning rate 0.03; 200 rounds; row and column sampling 0.8; minimum child weight 10	Maximum depth: 2, 3
Pooled Random Forest	As above	Minimum node size: 20
Pooled XGBoost	As above	Maximum depth: 3

4.11 Pooled models and one-hot encoding

Pooled Random Forest and XGBoost models combined training examples from USD, GBP and EUR. Currency identity was represented using three binary indicator columns: Currency_USD, Currency_GBP and Currency_EUR. These indicators enabled the models to distinguish observations from different currencies.

The pooled Random Forest used a minimum node size of twenty, while pooled XGBoost used a maximum depth of three. Both estimation-window approaches were evaluated. Since the currencies share an LKR component, pooled observations were not treated as independent sources of information.

Table 4.6: *One-hot encoding of currency identity*

Observation	Currency_USD	Currency_GBP	Currency_EUR
USD example	1	0	0
GBP example	0	1	0
EUR example	0	0	1

4.12 Parameter tuning and model selection

Candidate settings were evaluated during 2022–2023 using the walk-forward procedure. The experiment included 35 configurations across nine model families. For each currency and horizon, the configuration with the lowest validation RMSE was selected within each family. MAE and the configuration identifier were used to resolve ties.

The configuration with the lowest validation RMSE across families was then identified as the selected point-forecast model. Selected settings were saved before final testing. Candidates with failed forecasts were excluded from selection to avoid comparisons based on incomplete evaluation samples.

Table 4.7: *Candidate settings evaluated during validation*

Family	Specifications	Windows	Configurations
Naïve	Latest observed rate	—	1
ARIMA	(0,1,0), (0,1,0)+drift, (1,1,0), (0,1,1), (1,1,1)	Expanding, recent	10
ETS	SES; automatic non-seasonal ETS	Expanding, recent	4
Linear regression	Return-lag features; all features	Expanding, recent	4
VAR	Lag 1; lag 2	Expanding, recent	4
Random Forest	Minimum node size 5; 20	Expanding, recent	4
XGBoost	Maximum depth 2; 3	Expanding, recent	4
Pooled Random Forest	Minimum node size 20	Expanding, recent	2
Pooled XGBoost	Maximum depth 3	Expanding, recent	2
Total			**35**

Adapted from 03_candidate_settings.xlsx. The selected configurations are reported in Chapter 5.

4.13 Structural-change assessment

The validation period included the sharp exchange-rate movements of 2022. These observations were retained because they represent a relevant forecasting challenge. Validation errors were also summarised separately for 2022 and 2023 to examine differences across market conditions.

Comparing recent and expanding estimation windows provided an additional assessment of sensitivity to historical information. The procedure allowed models to adapt after new movements became observable but did not assume that unexpected shocks could be anticipated.

4.14 Volatility modelling and prediction intervals

Volatility was modelled separately from exchange-rate levels, using the same walk-forward origins. The volatility target was the next-reporting-day variance of percentage log returns, and the unobserved variance was proxied by the squared percentage log return realised at the target date.

4.14.1 Return input

Models were applied separately to the percentage log returns of each currency, $100 \times (\ln P_t - \ln P_{t-1})$, calculated between consecutive reporting days.

4.14.2 Model specification

Two volatility families were compared. The first was an exponentially weighted moving average (EWMA) of squared returns with decay parameters 0.94, 0.97 and 0.99 (RiskMetrics Group, 1996). The second was a zero-mean GARCH(1,1) model (Engle, 1982; Bollerslev, 1986) fitted with either normal or Student-t innovations. The Student-t option allows for the heavy tails identified in Chapter 3.

Table 4.8: *Volatility model candidates*

Family	Specification	Candidates
EWMA	Exponentially weighted variance	Decay 0.94, 0.97, 0.99
GARCH	Zero-mean standard GARCH(1,1)	Normal; Student-t innovations

4.14.3 Estimation and selection

Both families were estimated on a recent window of observations and re-estimated as the forecast origin moved forward. Within each family, the setting with the lowest validation QLIKE was selected for each currency, and the selected settings were frozen before testing. All candidate fits converged without failure in the validation and test samples (no fit failures were recorded).

4.14.4 Interval construction

Nominal 95% prediction intervals for exchange rates were generated from the selected GARCH models by simulation. Return paths were simulated over the five- and twenty-day horizons (2,000 paths per origin), accumulated into log changes, converted into exchange rates from the latest observed rate, and summarised by the simulated median and the 2.5th and 97.5th percentiles. These intervals belong to the zero-mean GARCH models and are not intervals around the separately selected point forecasts.

4.14.5 Evaluation

Volatility forecasts were evaluated with two loss functions, where σ^2 is the forecast variance and y^2 is the squared-return proxy: $\text{QLIKE} = \ln(\sigma^2) + y^2/\sigma^2$, and $\text{MSE} = (y^2 - \sigma^2)^2$. QLIKE is robust to noise in the proxy (Patton, 2011) and, because it is not bounded at zero, can take negative values. Lower values are better for both. Intervals were evaluated by their empirical coverage (the percentage of actual rates falling inside the interval) and mean width in LKR. Numerical results are presented in Chapter 5.

4.15 Forecast evaluation measures

Exchange-rate forecasts were assessed using MAE and RMSE, reported in LKR per unit of foreign currency. MAE measures the average absolute forecast error, while RMSE places greater weight on larger errors. Scores were calculated separately by currency and forecast horizon.

Forecast skill relative to the naïve model was also reported. Positive values indicated lower squared forecast errors than the benchmark, while negative values indicated worse performance. Level-based R-squared was treated as supplementary because trending exchange-rate levels can produce high values without strong forecasting skill.

Twenty-day forecasts issued every five observations overlap. Consequently, their errors are dependent, which must be considered when interpreting any statistical comparison.

Table 4.9: *Forecast evaluation measures*

Measure	Unit	Interpretation
MAE	LKR per unit	Average absolute error; lower is better
RMSE	LKR per unit	Penalises large errors; lower is better. Main selection criterion
RMSE improvement vs naïve	%	(Naïve RMSE − model RMSE) / naïve RMSE; positive = better than naïve
R^2 vs naïve	Ratio	1 − (model MSE / naïve MSE); positive = lower squared error than naïve
Level R^2	Ratio	Supplementary only; inflated by trending levels
QLIKE / MSE (volatility)	—	Loss against the squared-return proxy; lower is better
Coverage / mean width (intervals)	% / LKR	Share of actual rates inside the 95% interval / average interval width

4.16 Tuition-cost application

The application used illustrative tuition payments of 5,000 units of USD, GBP and EUR. Each forecast exchange rate was multiplied by the corresponding fixed fee to obtain a forecast payment cost in LKR. Where valid exchange-rate prediction intervals were available, their bounds were multiplied by the same positive fee amount to obtain tuition-cost intervals.

This application estimates the exchange-rate component of payment costs. It does not forecast changes in university fees and excludes bank spreads and transfer charges. Results are intended to illustrate budgeting uncertainty rather than identify a guaranteed cheapest payment date.

Table 4.10: *Assumed tuition fees*

Currency	Fee amount (foreign currency)
USD	5,000
GBP	5,000
EUR	5,000

4.17 Software and reproducibility

The analysis was implemented in R 4.6.1 using packages for data processing (dplyr, readr), statistical forecasting (forecast 9.0.2, vars, tseries, urca, lmtest, strucchange), machine learning (ranger 0.18.0, xgboost 3.2.1.1; Wright and Ziegler, 2017) and volatility modelling (rugarch 1.5-6). A base random seed of 2026 was specified, with reproducible origin-specific seeds used where implemented. Candidate settings and selected configurations were saved to support reproducibility. Tables were exported to Excel and figures to PNG files. Full package versions are listed in Appendix F.

Chapter 5: Results and Discussion

5.1 Overview of the evaluation

This chapter presents the validation and final testing results for exchange-rate forecasting and volatility modelling. Models were initially fitted using observations from 2000–2021, while 2022–2023 was used for parameter and estimation-window selection. Final performance was evaluated during 2024–2025 using the configurations selected during validation.

Results are reported separately for USD/LKR, GBP/LKR and EUR/LKR at five- and twenty-reporting-day horizons. Each model configuration was assessed at 97 test forecast origins per currency and horizon. No point-forecast fitting failures were recorded.

5.2 Training-period diagnostics and ARIMA candidates

Before fitting candidate models, diagnostics were run on the initial training period (2000–2021). Consistent with the full-sample results in Chapter 3, the augmented Dickey–Fuller test did not reject a unit root in log exchange-rate levels ($p = 0.60$ for USD, 0.47 for GBP and 0.43 for EUR), whereas KPSS rejected stationarity ($p \leq 0.01$). For log returns, ADF rejected a unit root ($p \leq 0.01$) and KPSS did not reject stationarity ($p \geq 0.10$). One difference was therefore recommended for all three series, supporting the ARIMA(p,1,q) specifications in Section 4.8.2. (The reported ADF and KPSS p-values are bounded by the tests' lookup tables.)

Table 5.1: *Training-period stationarity diagnostics (2000–2021)*

Currency	ADF p (log level)	KPSS p (log level)	ADF p (returns)	KPSS p (returns)	Differences (d)
USD	0.60	≤ 0.01	≤ 0.01	≥ 0.10	1
GBP	0.47	≤ 0.01	≤ 0.01	≥ 0.10	1
EUR	0.43	≤ 0.01	≤ 0.01	≥ 0.10	1

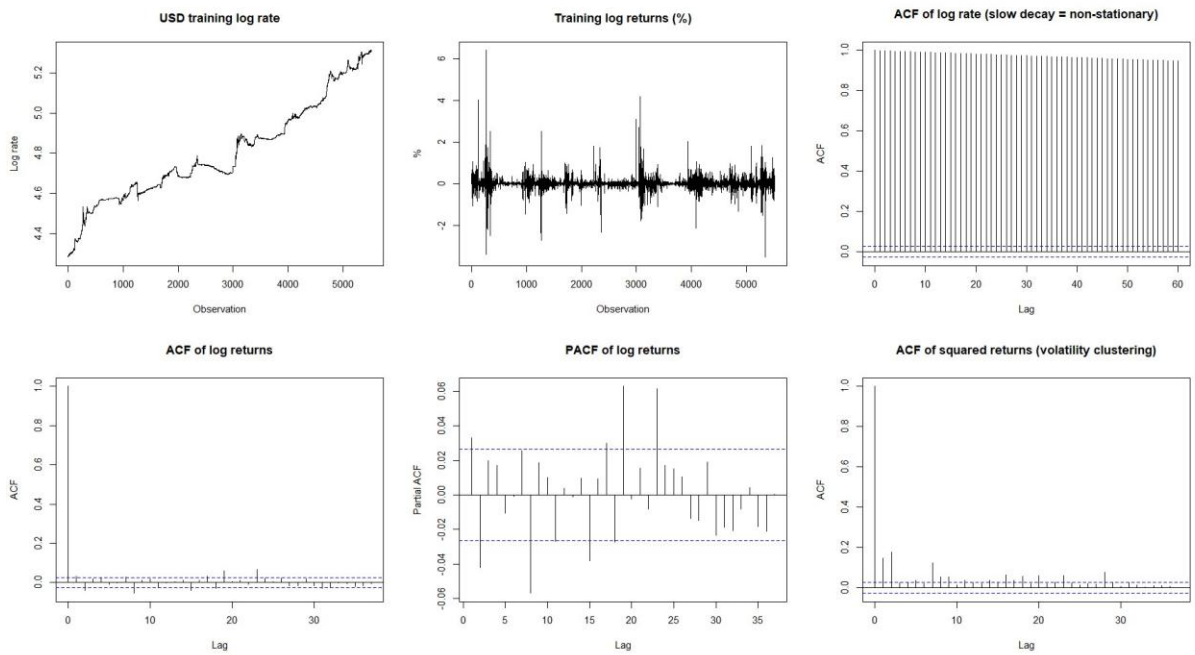


Figure 5.1: *USD/LKR training-period diagnostics: log rate, log returns and autocorrelation functions (2000–2021). GBP and EUR equivalents are in Appendix D.*

The autocorrelation function of the log rate decays very slowly, which is typical of a non-stationary series, whereas returns show little autocorrelation. Squared returns remain autocorrelated, indicating volatility clustering. Monthly seasonal strength was very small (0.007 for USD, 0.016 for GBP and 0.009 for EUR over 264 months), so seasonal components were not included in the candidate models.

Table 5.2 compares the ARIMA candidates on the training period. Drift-augmented ARIMA(0,1,0) had the lowest AIC for all three currencies, and the lowest BIC for USD, while ARIMA(0,1,0) without drift had the lowest BIC for GBP and EUR. The automatic search suggested ARIMA(2,1,0) with drift for USD and EUR and ARIMA(1,1,0) with drift for GBP, none of which was in the candidate list. Differences in information criteria between candidates were small for GBP and EUR. Ljung–Box tests at the tested lags indicated some remaining residual autocorrelation for every candidate, so information criteria alone could not settle the choice; as stated in Section 4.8.2, the final configuration was selected by validation RMSE.

Table 5.2: *ARIMA candidates on the training period (lowest AIC and BIC per currency in bold)*

Currency	Model	AIC	BIC	Ljung–Box p
USD	ARIMA(0,1,0)	-48,405.8	-48,399.2	< 0.0001
USD	ARIMA(0,1,0) + drift	**-48,425.2**	**-48,411.9**	< 0.0001
USD	ARIMA(1,1,0)	-48,411.3	-48,398.1	< 0.0001
USD	ARIMA(0,1,1)	-48,412.0	-48,398.7	< 0.0001
USD	ARIMA(1,1,1)	-48,416.9	-48,397.0	0.0002
GBP	ARIMA(0,1,0)	-39,642.8	**-39,636.2**	< 0.0001
GBP	ARIMA(0,1,0) + drift	**-39,643.7**	-39,630.5	< 0.0001
GBP	ARIMA(1,1,0)	-39,643.0	-39,629.8	< 0.0001
GBP	ARIMA(0,1,1)	-39,643.1	-39,629.8	< 0.0001
GBP	ARIMA(1,1,1)	-39,641.0	-39,621.2	< 0.0001
EUR	ARIMA(0,1,0)	-39,679.9	**-39,673.3**	0.0295
EUR	ARIMA(0,1,0) + drift	**-39,683.2**	-39,670.0	0.0295
EUR	ARIMA(1,1,0)	-39,677.9	-39,664.7	0.0178
EUR	ARIMA(0,1,1)	-39,677.9	-39,664.7	0.0177
EUR	ARIMA(1,1,1)	-39,675.9	-39,656.1	0.0102

Source: 02b_arima_aic_bic_ljungbox.xlsx. Auto-ARIMA cross-check: USD (2,1,0)+drift; GBP (1,1,0)+drift; EUR (2,1,0)+drift.

5.3 Models selected during validation

Validation results indicated that the preferred forecasting approach depended on the forecast horizon. Pooled Random Forest with an expanding estimation window was selected for five-day forecasts across all three currencies. For twenty-day forecasts, recent-window ETS was selected for GBP and EUR, while recent-window ARIMA(1,1,1) was selected for USD.

These configurations were fixed before final testing. Their selection reflected performance during 2022–2023 and did not guarantee that they would remain the best-performing models during 2024–2025.

Table 5.3: *Point-forecast models selected using validation RMSE (2022–2023)*

Currency	Horizon	Selected model	Window	Val. MAE	Val. RMSE	RMSE impr. vs naïve
USD	5 days	Pooled Random Forest	Expanding	3.921	7.636	7.51%
GBP	5 days	Pooled Random Forest	Expanding	7.509	11.568	3.50%
EUR	5 days	Pooled Random Forest	Expanding	6.046	9.605	2.99%
USD	20 days	ARIMA(1,1,1)	Recent 1,250 observations	11.696	20.328	6.84%
GBP	20 days	ETS	Recent 1,250 observations	18.699	27.505	1.92%
EUR	20 days	ETS	Recent 1,250 observations	15.467	23.246	2.97%

Source: 05_chosen_point_models.xlsx and 04_validation_scores.xlsx. MAE/RMSE in LKR per foreign-currency unit. The within-family selection table is in Appendix B.

5.4 Final test performance of the selected models

The validation-selected configurations were evaluated against the naïve benchmark on the final test period. The selected five-day pooled Random Forest models slightly improved RMSE for USD and EUR but performed worse than the benchmark for GBP. Improvements for USD and EUR were small, suggesting limited practical gains over assuming no change in the exchange rate.

At the twenty-day horizon, all three validation-selected models improved RMSE relative to the benchmark. The largest improvement was obtained for USD using ARIMA(1,1,1), which reduced RMSE by approximately 7.77%. The selected ETS models achieved smaller improvements for GBP and EUR.

Table 5.4: *Final test performance of validation-selected models (2024–2025)*

Currency	Horizon	Model	MAE	RMSE	Naïve RMSE	RMSE improvement
USD	5	Pooled RF	1.1353	1.6706	1.6751	0.27%
GBP	5	Pooled RF	3.5291	4.5327	4.3565	−4.04%
EUR	5	Pooled RF	2.8583	3.6850	3.6887	0.10%
USD	20	ARIMA(1,1,1)	2.6627	3.4980	3.7927	7.77%
GBP	20	ETS	7.1147	8.5528	8.6931	1.61%
EUR	20	ETS	6.2367	7.8823	8.0173	1.68%

MAE and RMSE are measured in LKR per foreign-currency unit. Negative improvement indicates worse performance than naïve. Source: 06_test_scores.xlsx.



Figure 5.2: *Final test forecasts of the validation-selected models compared with the naïve forecast and the actual rate, by currency and horizon (2024–2025). Plotted at the forecast target date.*

At the five-day horizon the selected model and the naïve forecast are almost indistinguishable from the actual rate at this scale, which is consistent with the very small RMSE differences in Table 5.4. At the twenty-day horizon the forecasts visibly lag turning points in the actual series, so errors are larger.

5.5 Comparison across all model families

Comparisons across all evaluated families showed that greater model complexity did not consistently improve forecasting performance. ARIMA produced the lowest observed five-day test RMSE for USD and GBP, while ETS produced the lowest five-day RMSE for EUR. At twenty days, ETS produced the lowest observed RMSE for USD and GBP, while linear regression produced the lowest RMSE for EUR.

These rankings describe the test results only. They were not used to replace the models selected before testing. Doing so would use the test set for model selection and would require a new evaluation period.

Table 5.5: *Lowest observed test RMSE across evaluated families*

Currency	Horizon	Lowest-RMSE family	RMSE	Improvement over naïve
USD	5	ARIMA	1.6377	2.23%
GBP	5	ARIMA	4.3548	0.04%
EUR	5	ETS	3.6739	0.40%
USD	20	ETS	3.4600	8.77%
GBP	20	ETS	8.5528	1.61%
EUR	20	Linear regression	7.7741	3.03%

Test-set ranking for description only; not used for model selection. Complete 54-row comparison: Appendix A.

Some differences were extremely small. For example, the five-day GBP improvement was approximately 0.04%.

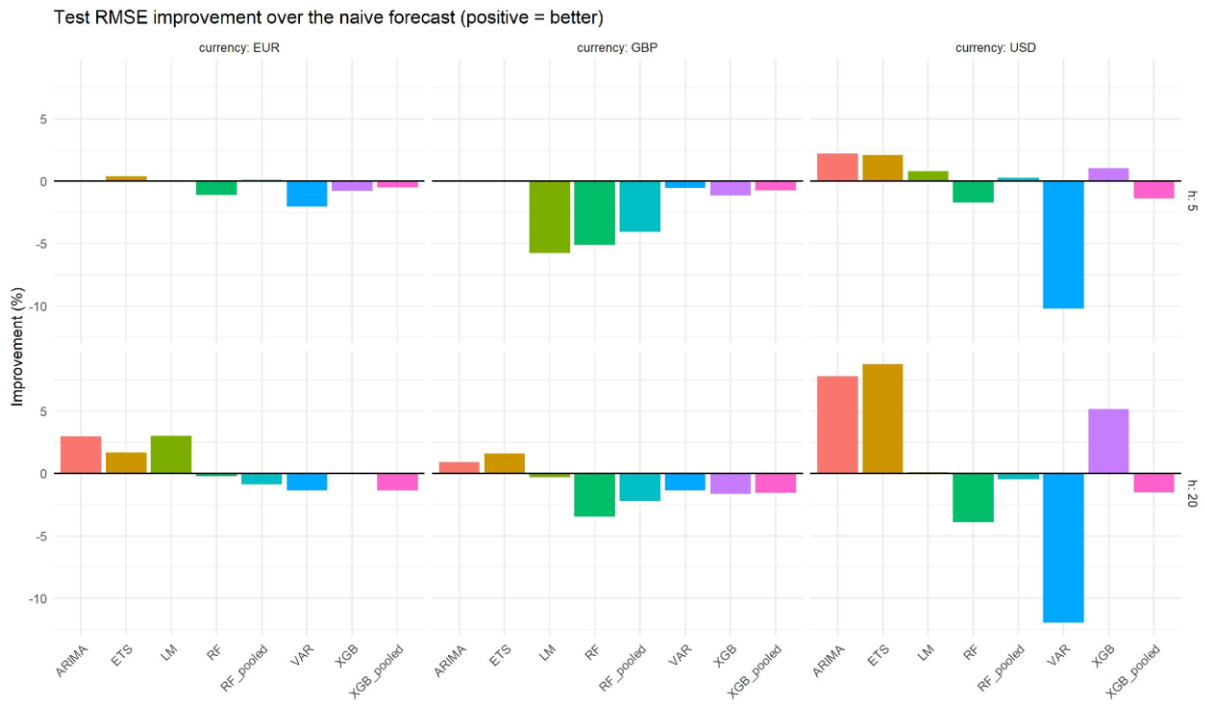


Figure 5.3: *Final test RMSE improvement relative to the naïve benchmark by currency and forecast horizon (positive = better than naïve).*

5.6 Forecast horizon and currency differences

Errors increased when the forecast horizon increased from five to twenty reporting days. This pattern was observed for the naïve benchmark and the selected models, indicating greater

uncertainty over longer forecasting horizons. For the naïve benchmark, test RMSE rose from 1.68 to 3.79 for USD, from 4.36 to 8.69 for GBP and from 3.69 to 8.02 for EUR, roughly a doubling in each case.

Absolute errors also differed across currencies. However, raw RMSE values should not be used alone to conclude that one currency is inherently easier to forecast because each exchange rate has a different numerical scale. Relative performance against the currency-specific naïve benchmark provides a more meaningful comparison. On that basis, the largest gains over naïve were obtained for USD at the twenty-day horizon, whereas GBP showed little or negative gain at five days.

Figure 5.2 shows only the validation-selected models. The complete forecast-versus-actual panels for every model family are provided in Appendix H.

5.7 Generalisation from validation to testing

Model performance did not transfer uniformly from validation to testing. Although pooled Random Forest was selected for all five-day forecasts during validation, its test-period improvements were negligible for USD and EUR and negative for GBP.

Table 5.6: *Validation RMSE of the selected models and the naïve benchmark by year*

Currency	Horizon	Model	2022 model	2022 naïve	2023 model	2023 naïve
USD	5	Pooled RF	7.80	8.40	7.46	8.09
GBP	5	Pooled RF	12.02	12.27	11.05	11.67
EUR	5	Pooled RF	9.82	10.04	9.36	9.74
USD	20	ARIMA(1,1,1)	24.58	27.80	15.16	13.85
GBP	20	ETS	33.48	34.74	20.18	19.58
EUR	20	ETS	27.29	29.04	18.55	17.76

RMSE in LKR per foreign-currency unit. Source: 04_validation_2022_vs_2023.xlsx.

Splitting the validation period by year shows that the pooled Random Forest had lower RMSE than the naïve forecast in both 2022 and 2023 for every currency at five days. At twenty days, the selected models beat the naïve forecast in 2022, but in 2023 the naïve forecast was slightly better for all three currencies. The validation advantage of the twenty-day models was therefore concentrated in the volatile 2022 period, yet in the final test period they still improved modestly on naïve. The five-day pooled Random Forest, by contrast, lost almost all of its validation advantage on the test data.

This result is consistent with sensitivity to changing market conditions and the limited predictability of exchange-rate movements. However, the difference cannot be attributed solely to the 2022 disruption without further analysis. Sampling variation, model specification and the limited hyperparameter search may also contribute.

The findings support retaining a naïve benchmark and evaluating models across different periods rather than assuming that the validation winner will always remain best.

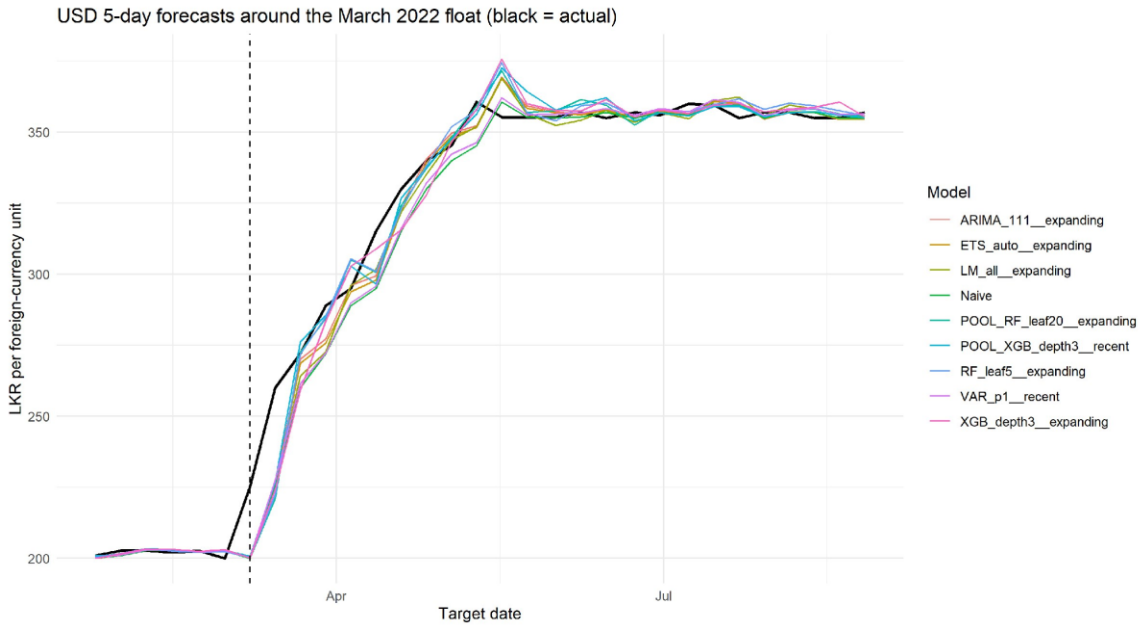


Figure 5.4: *Validation-period illustration (not a test result): five-day USD/LKR forecasts of the per-family validation winners around the March 2022 float (dashed line). Black = actual rate.*

In this illustration every model, including the naïve forecast, under-predicts during the rapid rise immediately after the float, because a shock cannot be anticipated from past data, and the forecasts converge on the actual rate once it stabilises.

5.8 Volatility forecasting results

Volatility forecasting was evaluated separately from exchange-rate level forecasting. EWMA and zero-mean GARCH(1,1) models predicted next-reporting-day variance using a recent estimation window. Squared percentage log returns were used as a noisy proxy for the unobserved variance.

Validation selected Student-t GARCH for USD and EUR and normal GARCH for GBP. The selected EWMA decay parameters were 0.94 for USD, 0.97 for GBP and 0.99 for EUR.

Table 5.7: *Final test volatility performance (97 test origins)*

Currency	Model	QLIKE	MSE
USD	EWMA, 0.94	−2.1628	**0.0152**
USD	Student-t GARCH	**−2.2049**	0.0190
GBP	EWMA, 0.97	**−0.5005**	**0.1773**
GBP	Normal GARCH	−0.4281	0.1949
EUR	EWMA, 0.99	**−0.6222**	**0.1516**
EUR	Student-t GARCH	−0.5885	0.1699

Lower scores are better. Bold values identify the better score within each currency. Source: 08_volatility_test_scores.xlsx.

EWMA outperformed GARCH for GBP and EUR under both evaluation measures. For USD, Student-t GARCH achieved lower QLIKE, while EWMA achieved lower MSE. Therefore, the results do not identify one universally superior volatility model.

Negative QLIKE values are valid under the loss definition used. Their interpretation depends on comparisons within the same currency and evaluation sample.

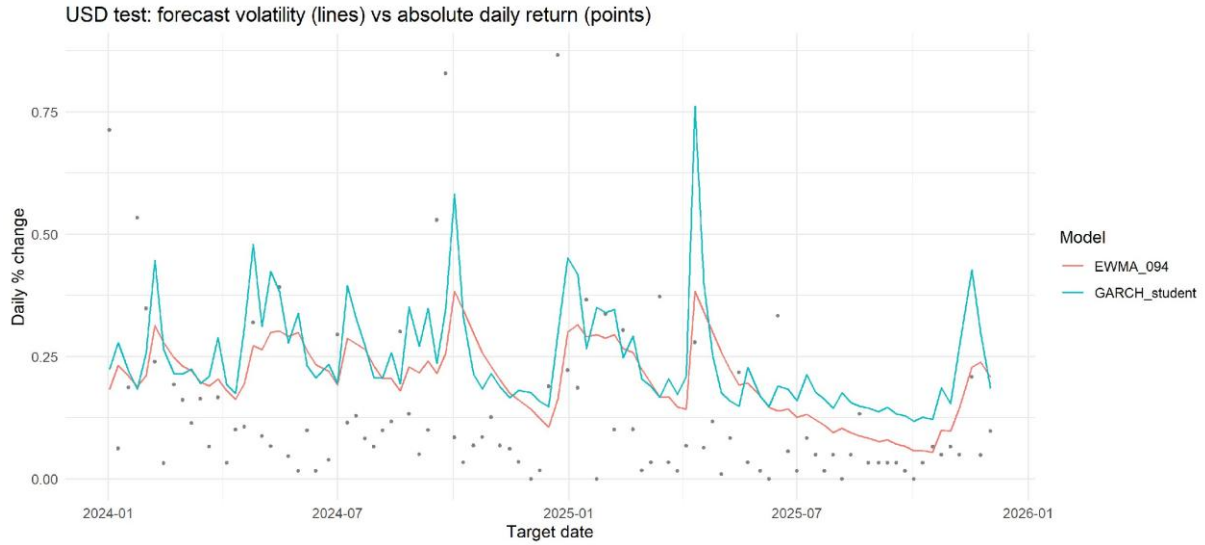


Figure 5.5: *USD/LKR: forecast conditional volatility (lines) compared with absolute observed daily returns (points) during the test period. GBP and EUR equivalents are in Appendix E.*

The plotted points are absolute returns, not directly observed true volatility.

5.9 Evaluation of GARCH prediction intervals

Separate GARCH simulations generated nominal 95% exchange-rate prediction intervals. Each interval was constructed from 2,000 simulated return paths. These intervals belong to the zero-mean GARCH models and should not be interpreted as intervals for the separately selected RF, ARIMA or ETS forecasts.

Table 5.8: *GARCH interval coverage and average width on the test period*

Currency	Horizon	GARCH distribution	Coverage	Average width
USD	5	Student-t	94.85%	6.4926
USD	20	Student-t	90.72%	14.7910
GBP	5	Normal	98.97%	20.3211
GBP	20	Normal	98.97%	47.8661
EUR	5	Student-t	96.91%	17.3111
EUR	20	Student-t	96.91%	38.1807

Widths are measured in LKR per foreign-currency unit. Nominal level 95%; 97 origins. Source: 09_garch_interval_test_scores.xlsx.

USD five-day coverage was close to the nominal 95% level, whereas twenty-day coverage was lower at 90.72%. GBP and EUR coverage exceeded 95%, which may indicate conservative intervals over this sample. Coverage must be assessed alongside width because wider intervals can achieve greater coverage while providing less precise guidance.

The twenty-day intervals were wider for every currency. Coverage estimates were based on 97 origins, with overlapping forecast periods, so they should not be interpreted as precise guarantees of future reliability.

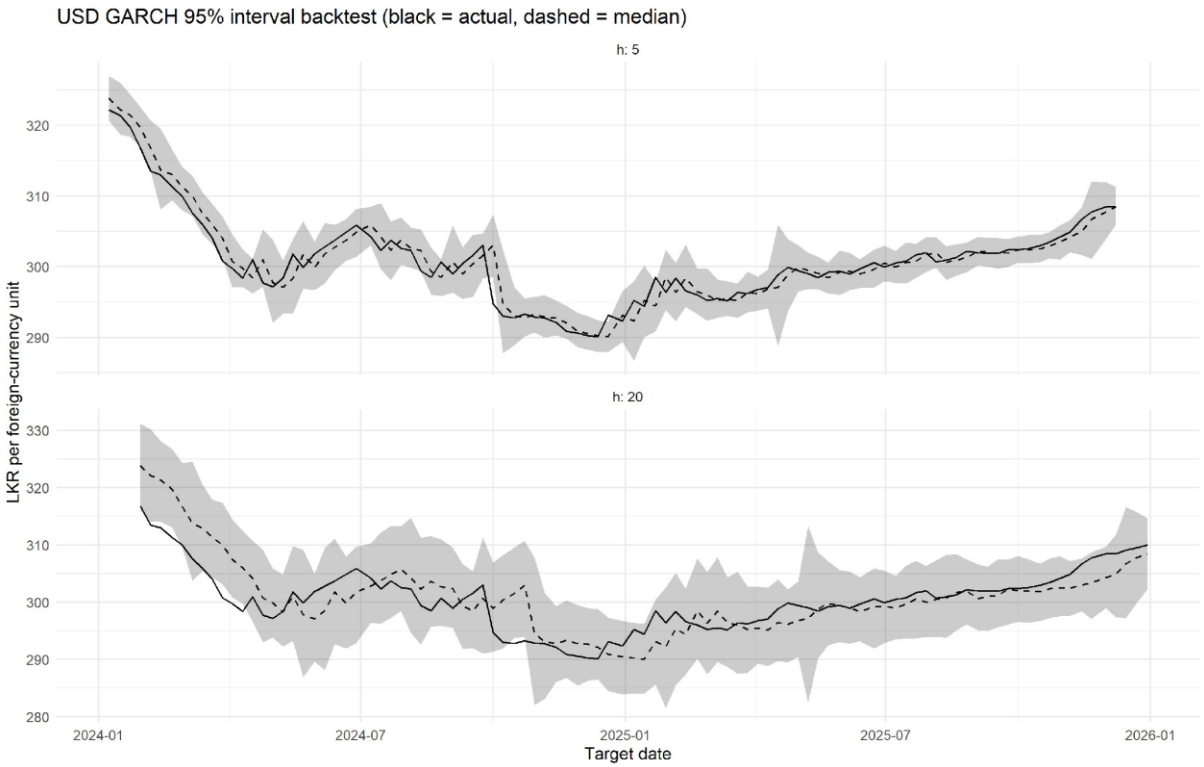


Figure 5.6: *USD/LKR GARCH 95% interval backtest on the test period (black = actual, dashed = simulated median, shaded = interval). GBP and EUR equivalents are in Appendix E.*

5.10 Final residual diagnostics

Residual diagnostics were conducted on models refitted through the end of 2025. ARIMA residuals showed significant remaining autocorrelation and ARCH effects at the tested lags. Consequently, their residuals could not be described as fully independent with constant variance.

For GARCH-standardised residuals, the ARCH-LM tests did not reject the absence of remaining ARCH effects. However, the USD Ljung–Box test remained significant, suggesting residual serial dependence. GBP and EUR did not show significant autocorrelation under the reported Ljung–Box test.

Table 5.9: *Final residual diagnostics (models refitted through 31 December 2025)*

Currency	Model	Ljung–Box p-value	ARCH-LM p-value
USD	ARIMA(1,1,1), expanding	< 0.0001	< 0.0001
USD	ARIMA(1,1,1), recent	< 0.0001	< 0.0001
USD	GARCH, Student-t	0.0147	0.9996
GBP	ARIMA(0,1,1), expanding	0.0007	< 0.0001
GBP	ARIMA(1,1,1), recent	0.0002	< 0.0001
GBP	GARCH, normal	0.1874	0.9964
EUR	ARIMA(0,1,1), recent	< 0.0001	< 0.0001
EUR	ARIMA(1,1,1), recent	0.0008	< 0.0001
EUR	GARCH, Student-t	0.3826	0.9897

Rows for GARCH are on standardised residuals. Source: 11_final_residual_checks.xlsx.

These checks describe the final fitted models. They do not independently establish out-of-sample forecast quality.

5.11 Summary of findings

The results show that forecasting performance depended on currency, horizon and evaluation period. Machine-learning models did not consistently outperform simpler alternatives. Selected twenty-day statistical models provided modest improvements over the naïve benchmark, while five-day pooled Random Forest performance was mixed.

Volatility modelling provided additional information about uncertainty, but GARCH did not consistently outperform EWMA. Prediction intervals generally widened with the horizon, and USD twenty-day undercoverage demonstrated the importance of evaluating uncertainty estimates rather than relying on their nominal confidence level.

Chapter 6: Tuition-Cost Application

6.1 Application assumptions

The application considered hypothetical tuition payments of USD 5,000, GBP 5,000 and EUR 5,000. Foreign-currency fee amounts were held constant, allowing the analysis to isolate exchange-rate effects on payment costs.

After evaluation, models were refitted using observations available through 31 December 2025. Forecasts were generated five and twenty reporting days beyond that origin. These are historical-origin scenarios rather than current payment forecasts. Exact calendar payment dates were not assigned because future reporting holidays were not specified.

Table 6.1: *Assumed tuition fees*

Currency	Fee amount (foreign currency)
USD	5,000
GBP	5,000
EUR	5,000

6.2 Point estimates of tuition costs

Tuition-cost point estimates were obtained by multiplying the fixed fee by the exchange-rate forecast from the validation-selected model. The latest-rate cost was included as a reference showing the amount required at the forecast origin.

Table 6.2: *Tuition-cost point forecasts as of 31 December 2025*

Currency	Horizon	Latest-rate cost, LKR	Forecast cost, LKR	Difference, LKR
USD	5	1,547,000	1,548,710	+1,710
USD	20	1,547,000	1,550,068	+3,068
GBP	5	2,080,251	2,082,350	+2,099
GBP	20	2,080,251	2,078,015	−2,236
EUR	5	1,815,559	1,818,988	+3,429
EUR	20	1,815,559	1,813,198	−2,361

Values are rounded to the nearest rupee. Source: 10_tuition_point_forecasts.xlsx.

The forecast changes were relatively small compared with the total payment amounts. However, these estimates are uncertain and do not demonstrate that delaying or advancing payment would produce savings.

6.3 GARCH tuition-cost scenarios

GARCH-based tuition scenarios were reported separately using simulated median costs and central 95% intervals. The interval bounds were converted from exchange rates into LKR costs using the fixed tuition amounts. These scenarios were not centred on the separately selected point forecasts.

Table 6.3: *GARCH tuition-cost scenarios as of 31 December 2025*

Currency	Horizon	Median cost, LKR	Lower bound, LKR	Upper bound, LKR
USD	5	1,546,936	1,535,036	1,558,261
USD	20	1,546,893	1,518,146	1,573,897
GBP	5	2,079,253	2,040,753	2,121,969
GBP	20	2,081,958	1,987,579	2,182,600
EUR	5	1,814,907	1,776,404	1,850,885
EUR	20	1,815,247	1,727,715	1,907,504

Nominal central 95% intervals. Source: 10_garch_tuition_scenarios.xlsx.

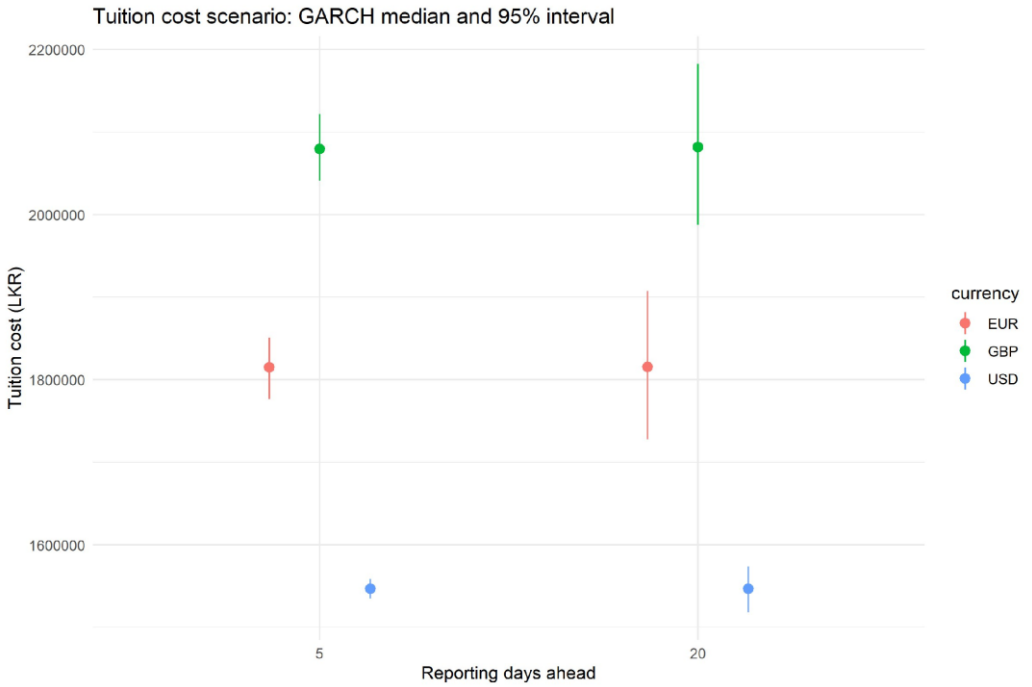


Figure 6.1: *Simulated GARCH median tuition costs and nominal 95% prediction intervals, based on information through 31 December 2025.*

The GBP payment carries the widest interval in rupee terms, and for every currency the twenty-day interval is wider than the five-day interval.

6.4 Illustrative planning buffers

An illustrative planning buffer was calculated as the difference between the GARCH median cost and the upper interval bound. This expresses the upper-side uncertainty in monetary terms rather than presenting only an exchange-rate range.

Table 6.4: *Illustrative planning buffers (upper bound minus median)*

Currency	Five-day buffer, LKR	Twenty-day buffer, LKR
USD	11,325	27,004
GBP	42,716	100,643
EUR	35,977	92,257

For example, the twenty-day GBP scenario placed the median payment cost at approximately LKR 2.082 million and the upper bound at LKR 2.183 million, a difference of approximately LKR 100,643.

These amounts are model-based illustrations rather than guaranteed sufficient reserves. In particular, the USD twenty-day intervals achieved only 90.72% test coverage, despite their nominal 95% level. The buffers should not be read as optimal or guaranteed amounts.

6.5 Interactive tuition-cost dashboard

An interactive dashboard was developed using R Shiny to present the forecasting results in a format suitable for tuition-payment planning. Users can select USD, GBP or EUR, choose a five- or twenty-reporting-day forecast horizon, and enter a foreign-currency tuition amount. The dashboard calculates the estimated payment cost in Sri Lankan rupees using the corresponding validation-selected exchange-rate forecast.

A horizontal range bar displays the separate GARCH-based central 95% exchange-rate scenario, together with markers for the latest observed rate and the point forecast. These values help users compare the forecast with the rate available at the forecast origin and understand the uncertainty represented by the GARCH scenario. The GARCH bounds are not prediction intervals for the separately selected point-forecast model.

The dashboard uses saved model outputs based on information available through 31 December 2025. Changing the tuition amount updates the displayed cost without retraining the models. It therefore serves as an illustrative budgeting tool rather than a live exchange-rate service. Bank spreads and transfer charges are excluded.



Figure 6.2: Interactive tuition-cost dashboard showing an illustrative USD 5,000 payment at the five-reporting-day horizon, based on forecasts as of 31 December 2025.

6.6 Practical interpretation

The application demonstrates that exchange-rate uncertainty can create meaningful variation in tuition-payment requirements. A point forecast provides a central planning estimate, while a prediction interval communicates a broader range of possible outcomes.

Students and families could use such outputs to assess whether their available funds cover a range of plausible costs. However, the results do not establish an optimal payment date. Actual payment decisions must also consider deadlines, bank selling rates, transfer charges and the consequences of late payment.

6.7 Application boundaries

The scenarios exclude tuition-fee increases, living expenses and transaction charges. GBP/LKR and EUR/LKR are derived cross-rates, while the source observations are reference rates rather than executable bank quotations.

The GARCH intervals are conditional on estimated parameters and modelling assumptions. They do not fully account for parameter uncertainty or unforeseen changes in exchange-rate regimes. No comparison with actual student transactions was conducted, so financial savings were not demonstrated.

Chapter 7: Conclusion, Limitations, and Recommendations

7.1 Conclusion

This study compared statistical and machine-learning methods for forecasting exchange rates relevant to overseas tuition payments by Sri Lankan students. Daily observations from 2000–2025 were analysed using chronological validation and walk-forward testing.

No model consistently dominated across currencies and forecast horizons. Pooled Random Forest was selected for five-day forecasts during validation, but its test-period performance was only marginally better than naïve for USD and EUR and worse for GBP. The selected twenty-day ARIMA and ETS models improved on the naïve benchmark, although improvements varied by currency.

These findings indicate that complex models should be evaluated against simple benchmarks rather than assumed to provide superior forecasts. In answer to the research question, exchange rates were only modestly forecastable: gains over the naïve benchmark were small at five days and moderate at twenty days (1.6%–7.8% for the selected models), and forecasts were most useful when presented together with their uncertainty.

7.2 Conclusions on volatility and uncertainty

Volatility modelling added an uncertainty perspective to the price forecasts. EWMA performed better than GARCH for GBP and EUR under both reported loss measures. For USD, the preferred model depended on the evaluation metric.

GARCH simulations produced tuition-cost ranges that widened with the forecast horizon. However, interval performance differed across currencies. The undercoverage of USD twenty-day intervals showed that nominal confidence levels should not be treated as guarantees.

7.3 Contribution to tuition-payment planning

The main practical contribution was translating exchange-rate forecasts and uncertainty into understandable LKR payment scenarios. This demonstrated how even a fixed foreign-currency fee can result in uncertain rupee funding requirements.

The application supports financial awareness and budgeting discussions. It does not demonstrate realised savings or guarantee the cheapest payment date.

7.4 Limitations

Several limitations should be considered when interpreting the findings.

First, the three currencies share an LKR component, and GBP/LKR and EUR/LKR were derived using cross-rate multiplication. Their relationships therefore partly reflect their construction.

Second, exchange-rate behaviour changed substantially during the study period. Walk-forward refitting and recent windows allowed adaptation after observations became available but could not anticipate unexpected structural changes.

Third, the machine-learning parameter grid was limited. The findings apply to the tested configurations rather than every possible implementation of the model families.

Fourth, the evaluation used 97 test origins per currency and horizon. Twenty-day forecast periods overlapped, creating dependence between errors and limiting simple significance claims.

Fifth, volatility was evaluated against squared returns, which are noisy proxies for unobserved variance. The GARCH models assumed a zero conditional mean, and USD-standardised residuals retained significant autocorrelation.

Finally, prediction intervals omitted parameter uncertainty and unforeseen regimes. The tuition application used hypothetical fees and excluded bank spreads, charges and changes in university fees.

7.5 Recommendations

Future implementations should retain the naïve benchmark and monitor model performance over time. Models should be periodically refitted using newly available observations, with the updating procedure specified before evaluation.

Any revision to model selection after reviewing these test results should be assessed on a new, untouched period. Final test rankings should not be reused as evidence that a newly chosen model will perform best in future.

Future research could investigate broader but controlled parameter searches, alternative conditional-mean specifications, asymmetric volatility models and interval calibration. Additional predictors should be included only when their historical availability at each forecast origin can be established.

For a practical student-facing application, recent data, executable bank selling rates and transfer charges should be incorporated. Model outputs should remain clearly labelled as uncertain estimates.

References

Board of Governors of the Federal Reserve System (US). (n.d.-a). *Sri Lankan rupees to U.S. dollar spot exchange rate [DEXSLUS]* [Data set]. FRED, Federal Reserve Bank of St. Louis. https://fred.stlouisfed.org/series/DEXSLUS

Board of Governors of the Federal Reserve System (US). (n.d.-b). *U.S. dollars to euro spot exchange rate [DEXUSEU]* [Data set]. FRED, Federal Reserve Bank of St. Louis. https://fred.stlouisfed.org/series/DEXUSEU

Board of Governors of the Federal Reserve System (US). (n.d.-c). *U.S. dollars to U.K. pound sterling spot exchange rate [DEXUSUK]* [Data set]. FRED, Federal Reserve Bank of St. Louis. https://fred.stlouisfed.org/series/DEXUSUK

Appendices

The appendices contain supplementary tables and figures that support the main chapters.

Appendix A: Complete test-set comparison of selected configurations

Test scores of the best validation configuration within each family, for every currency and horizon (54 rows). RMSE improvement is relative to the naïve benchmark.

Table 5: *Complete test comparison across all families*

Currency	h	Family	Configuration	MAE	RMSE	RMSE impr. %
USD	5	ARIMA	ARIMA_111 / expanding	1.1122	1.6377	2.23
USD	5	ETS	ETS_auto / expanding	1.1150	1.6399	2.10
USD	5	XGBoost	XGB_depth3 / expanding	1.1531	1.6580	1.02
USD	5	Linear regression	LM_all / expanding	1.1373	1.6612	0.83
USD	5	Pooled RF	POOL_RF_leaf20 / expanding	1.1353	1.6706	0.27
USD	5	Naive	Naive	1.2130	1.6751	0.00
USD	5	Pooled XGBoost	POOL_XGB_depth3 / recent	1.1992	1.6979	−1.36
USD	5	Random Forest	RF_leaf5 / expanding	1.1465	1.7040	−1.73
USD	5	VAR	VAR_p1 / recent	1.3367	1.8465	−10.23
USD	20	ETS	ETS_auto / expanding	2.6620	3.4600	8.77
USD	20	ARIMA	ARIMA_111 / recent	2.6627	3.4980	7.77
USD	20	XGBoost	XGB_depth2 / expanding	2.6812	3.5970	5.16
USD	20	Linear regression	LM_all / expanding	2.7777	3.7895	0.08
USD	20	Naive	Naive	2.8998	3.7927	0.00
USD	20	Pooled RF	POOL_RF_leaf20 / expanding	2.9161	3.8094	−0.44
USD	20	Pooled XGBoost	POOL_XGB_depth3 / expanding	2.8586	3.8495	−1.50
USD	20	Random Forest	RF_leaf20 / expanding	2.9977	3.9409	−3.91
USD	20	VAR	VAR_p1 / expanding	3.0160	4.2465	−11.96
GBP	5	ARIMA	ARIMA_011 / expanding	3.3962	4.3548	0.04
GBP	5	Naive	Naive	3.3942	4.3565	0.00
GBP	5	ETS	SES / expanding	3.3942	4.3565	−0.00
GBP	5	VAR	VAR_p1 / expanding	3.4288	4.3804	−0.55
GBP	5	Pooled XGBoost	POOL_XGB_depth3 / expanding	3.4409	4.3897	−0.76
GBP	5	XGBoost	XGB_depth3 / expanding	3.4558	4.4055	−1.13
GBP	5	Pooled RF	POOL_RF_leaf20 / expanding	3.5291	4.5327	−4.04
GBP	5	Random Forest	RF_leaf5 / expanding	3.6350	4.5794	−5.12
GBP	5	Linear regression	LM_all / recent	3.6125	4.6063	−5.73
GBP	20	ETS	ETS_auto / recent	7.1147	8.5528	1.61
GBP	20	ARIMA	ARIMA_111 / recent	7.0082	8.6116	0.94
GBP	20	Naive	Naive	7.3734	8.6931	0.00
GBP	20	Linear regression	LM_lags / expanding	7.1893	8.7177	−0.28
GBP	20	VAR	VAR_p1 / expanding	7.3176	8.8110	−1.36
GBP	20	Pooled XGBoost	POOL_XGB_depth3 / expanding	7.2986	8.8289	−1.56

Currency	h	Family	Configuration	MAE	RMSE	RMSE impr. %
GBP	20	XGBoost	XGB_depth2 / expanding	7.3750	8.8361	−1.64
GBP	20	Pooled RF	POOL_RF_leaf20 / expanding	7.3262	8.8849	−2.21
GBP	20	Random Forest	RF_leaf5 / expanding	7.4469	8.9943	−3.46
EUR	5	ETS	ETS_auto / expanding	2.8555	3.6739	0.40
EUR	5	Pooled RF	POOL_RF_leaf20 / expanding	2.8583	3.6850	0.10
EUR	5	ARIMA	ARIMA_011 / recent	2.8831	3.6880	0.02
EUR	5	Naive	Naive	2.8817	3.6887	0.00
EUR	5	Linear regression	LM_all / expanding	2.8816	3.6889	−0.00
EUR	5	Pooled XGBoost	POOL_XGB_depth3 / expanding	2.9278	3.7068	−0.49
EUR	5	XGBoost	XGB_depth3 / expanding	2.9190	3.7174	−0.78
EUR	5	Random Forest	RF_leaf20 / expanding	2.8919	3.7296	−1.11
EUR	5	VAR	VAR_p1 / recent	2.9711	3.7635	−2.03
EUR	20	Linear regression	LM_all / expanding	6.2231	7.7741	3.03
EUR	20	ARIMA	ARIMA_111 / recent	6.1719	7.7787	2.98
EUR	20	ETS	ETS_auto / recent	6.2367	7.8823	1.68
EUR	20	Naive	Naive	6.3846	8.0173	0.00
EUR	20	XGBoost	XGB_depth2 / expanding	6.2945	8.0183	−0.01
EUR	20	Random Forest	RF_leaf20 / expanding	6.4393	8.0334	−0.20
EUR	20	Pooled RF	POOL_RF_leaf20 / expanding	6.4448	8.0866	−0.86
EUR	20	VAR	VAR_p1 / expanding	6.3698	8.1238	−1.33
EUR	20	Pooled XGBoost	POOL_XGB_depth3 / expanding	6.4739	8.1247	−1.34

Appendix B: Within-family validation selection

Table 6: *Best configuration per family on validation RMSE (2022–2023)*

Currency	h	Family	Configuration	MAE	RMSE	R^2 vs naïve
USD	5	Pooled RF	POOL_RF_leaf20 / expanding	3.921	7.636	0.1445
USD	5	Random Forest	RF_leaf5 / expanding	4.163	7.948	0.0732
USD	5	ARIMA	ARIMA_111 / expanding	4.174	7.962	0.0700
USD	5	Pooled XGBoost	POOL_XGB_depth3 / recent	4.413	8.075	0.0434
USD	5	ETS	ETS_auto / expanding	4.235	8.085	0.0410
USD	5	VAR	VAR_p1 / recent	4.513	8.170	0.0207
USD	5	Linear regression	LM_all / expanding	4.566	8.197	0.0142
USD	5	Naive	Naive	4.397	8.256	0.0000
USD	5	XGBoost	XGB_depth3 / expanding	4.464	8.271	−0.0036
USD	20	ARIMA	ARIMA_111 / recent	11.696	20.328	0.1321
USD	20	ETS	ETS_auto / expanding	11.749	20.903	0.0824
USD	20	Linear regression	LM_all / expanding	12.941	21.479	0.0311
USD	20	VAR	VAR_p1 / expanding	11.499	21.629	0.0175
USD	20	Naive	Naive	11.480	21.821	0.0000
USD	20	XGBoost	XGB_depth2 / expanding	13.961	22.937	−0.1050
USD	20	Random Forest	RF_leaf20 / expanding	15.075	23.545	−0.1643
USD	20	Pooled XGBoost	POOL_XGB_depth3 / expanding	13.282	23.840	−0.1937

Currency	h	Family	Configuration	MAE	RMSE	R^2 vs naïve
USD	20	Pooled RF	POOL_RF_leaf20 / expanding	13.782	24.040	−0.2138
GBP	5	Pooled RF	POOL_RF_leaf20 / expanding	7.509	11.568	0.0688
GBP	5	XGBoost	XGB_depth3 / expanding	7.434	11.658	0.0543
GBP	5	Random Forest	RF_leaf5 / expanding	7.575	11.684	0.0500
GBP	5	VAR	VAR_p1 / expanding	7.641	11.906	0.0136
GBP	5	ARIMA	ARIMA_011 / expanding	7.671	11.960	0.0046
GBP	5	Naive	Naive	7.681	11.988	0.0000
GBP	5	ETS	SES / expanding	7.681	11.988	−0.0000
GBP	5	Linear regression	LM_all / recent	8.095	12.035	−0.0079
GBP	5	Pooled XGBoost	POOL_XGB_depth3 / expanding	7.830	12.070	−0.0137
GBP	20	ETS	ETS_auto / recent	18.699	27.505	0.0381
GBP	20	ARIMA	ARIMA_111 / recent	18.573	27.726	0.0226
GBP	20	VAR	VAR_p1 / expanding	18.816	27.849	0.0140
GBP	20	Naive	Naive	18.758	28.045	0.0000
GBP	20	Linear regression	LM_lags / expanding	18.974	28.129	−0.0060
GBP	20	XGBoost	XGB_depth2 / expanding	19.999	29.172	−0.0819
GBP	20	Random Forest	RF_leaf5 / expanding	21.369	30.325	−0.1692
GBP	20	Pooled RF	POOL_RF_leaf20 / expanding	21.413	30.879	−0.2123
GBP	20	Pooled XGBoost	POOL_XGB_depth3 / expanding	20.920	30.880	−0.2124
EUR	5	Pooled RF	POOL_RF_leaf20 / expanding	6.046	9.605	0.0590
EUR	5	Random Forest	RF_leaf20 / expanding	6.034	9.670	0.0463
EUR	5	Linear regression	LM_all / expanding	6.255	9.769	0.0266
EUR	5	VAR	VAR_p1 / recent	6.247	9.811	0.0182
EUR	5	Pooled XGBoost	POOL_XGB_depth3 / expanding	6.181	9.839	0.0126
EUR	5	ARIMA	ARIMA_011 / recent	6.271	9.850	0.0104
EUR	5	ETS	ETS_auto / expanding	6.302	9.870	0.0062
EUR	5	Naive	Naive	6.306	9.901	0.0000
EUR	5	XGBoost	XGB_depth3 / expanding	6.499	10.153	−0.0516
EUR	20	ETS	ETS_auto / recent	15.467	23.246	0.0586
EUR	20	Linear regression	LM_all / expanding	15.754	23.324	0.0522
EUR	20	ARIMA	ARIMA_111 / recent	15.738	23.636	0.0267
EUR	20	VAR	VAR_p1 / expanding	15.770	23.779	0.0149
EUR	20	Naive	Naive	15.812	23.958	0.0000
EUR	20	Pooled XGBoost	POOL_XGB_depth3 / expanding	16.904	25.705	−0.1512
EUR	20	Pooled RF	POOL_RF_leaf20 / expanding	17.492	26.018	−0.1793
EUR	20	XGBoost	XGB_depth2 / expanding	17.530	26.431	−0.2171
EUR	20	Random Forest	RF_leaf20 / expanding	18.107	26.434	−0.2174

Appendix C: Full candidate-settings table

Table 7: *All 35 candidate configurations*

Identifier	Family	Specification	Window
Naive	Naïve	—	expanding
ARIMA_010__expanding	ARIMA	order (0,1,0)	expanding

Identifier	Family	Specification	Window
ARIMA_010_drift__expanding	ARIMA	order (0,1,0) + drift	expanding
ARIMA_110__expanding	ARIMA	order (1,1,0)	expanding
ARIMA_011__expanding	ARIMA	order (0,1,1)	expanding
ARIMA_111__expanding	ARIMA	order (1,1,1)	expanding
RF_leaf5__expanding	Random Forest	min node size 5	expanding
RF_leaf20__expanding	Random Forest	min node size 20	expanding
XGB_depth2__expanding	XGBoost	max depth 2	expanding
XGB_depth3__expanding	XGBoost	max depth 3	expanding
SES__expanding	ETS	ETS model ANN	expanding
ETS_auto__expanding	ETS	ETS model ZZN	expanding
LM_all__expanding	Linear regression	features: all	expanding
LM_lags__expanding	Linear regression	features: lags	expanding
VAR_p1__expanding	VAR	VAR lag 1	expanding
VAR_p2__expanding	VAR	VAR lag 2	expanding
POOL_RF_leaf20__expanding	Pooled RF	min node size 20	expanding
POOL_XGB_depth3__expanding	Pooled XGBoost	max depth 3	expanding
ARIMA_010__recent	ARIMA	order (0,1,0)	recent
ARIMA_010_drift__recent	ARIMA	order (0,1,0) + drift	recent
ARIMA_110__recent	ARIMA	order (1,1,0)	recent
ARIMA_011__recent	ARIMA	order (0,1,1)	recent
ARIMA_111__recent	ARIMA	order (1,1,1)	recent
RF_leaf5__recent	Random Forest	min node size 5	recent
RF_leaf20__recent	Random Forest	min node size 20	recent
XGB_depth2__recent	XGBoost	max depth 2	recent
XGB_depth3__recent	XGBoost	max depth 3	recent
SES__recent	ETS	ETS model ANN	recent
ETS_auto__recent	ETS	ETS model ZZN	recent
LM_all__recent	Linear regression	features: all	recent
LM_lags__recent	Linear regression	features: lags	recent
VAR_p1__recent	VAR	VAR lag 1	recent
VAR_p2__recent	VAR	VAR lag 2	recent
POOL_RF_leaf20__recent	Pooled RF	min node size 20	recent
POOL_XGB_depth3__recent	Pooled XGBoost	max depth 3	recent

Appendix D: Supplementary training-period diagnostics

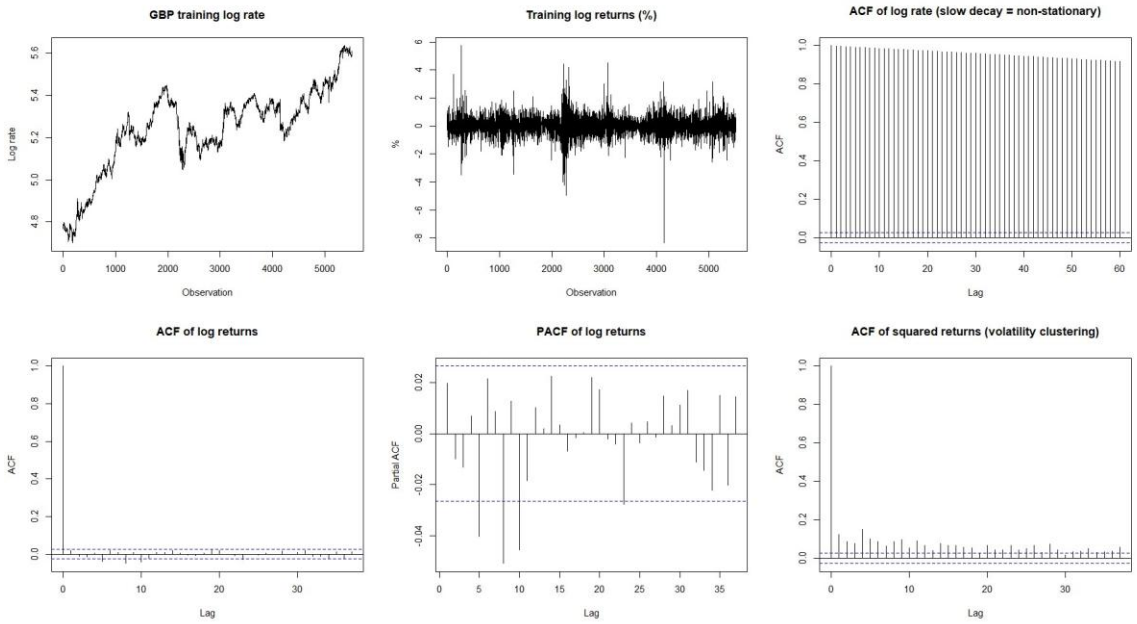


Figure 2: *GBP/LKR training-period diagnostics (2000–2021).*

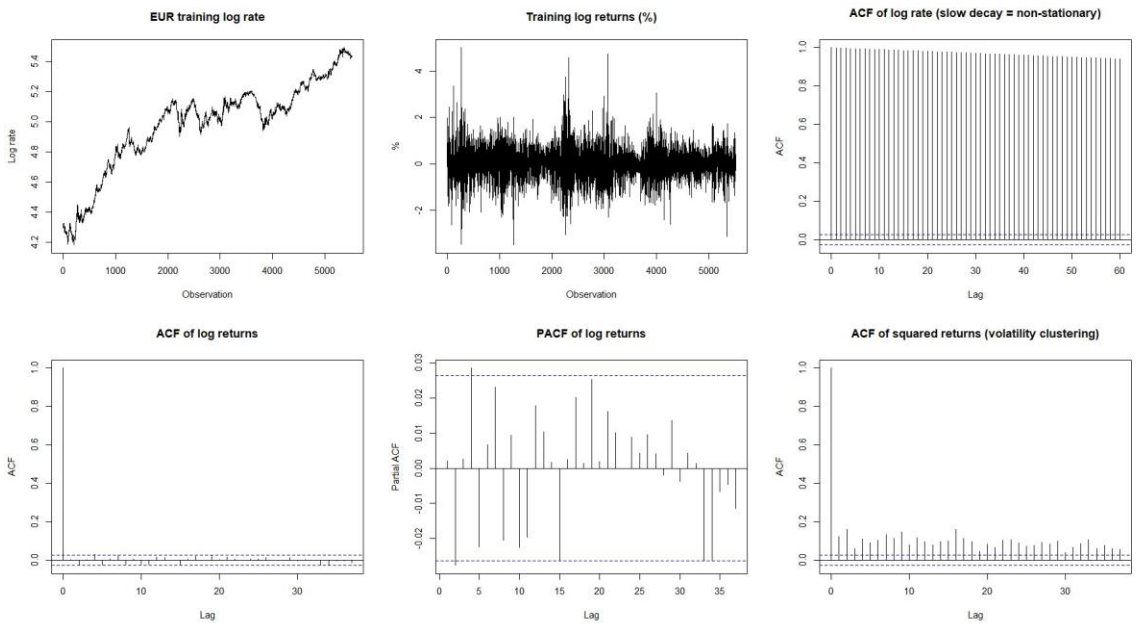


Figure 3: *EUR/LKR training-period diagnostics (2000–2021).*

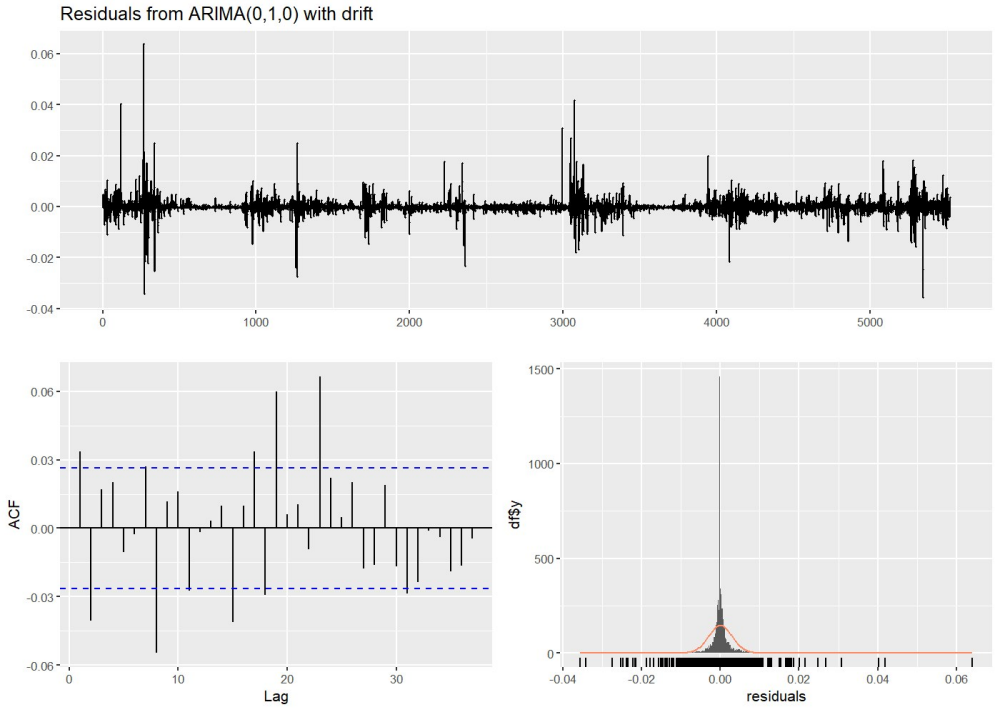


Figure 4: *Residual diagnostics of ARIMA(0,1,0) with drift, USD/LKR (training period).*

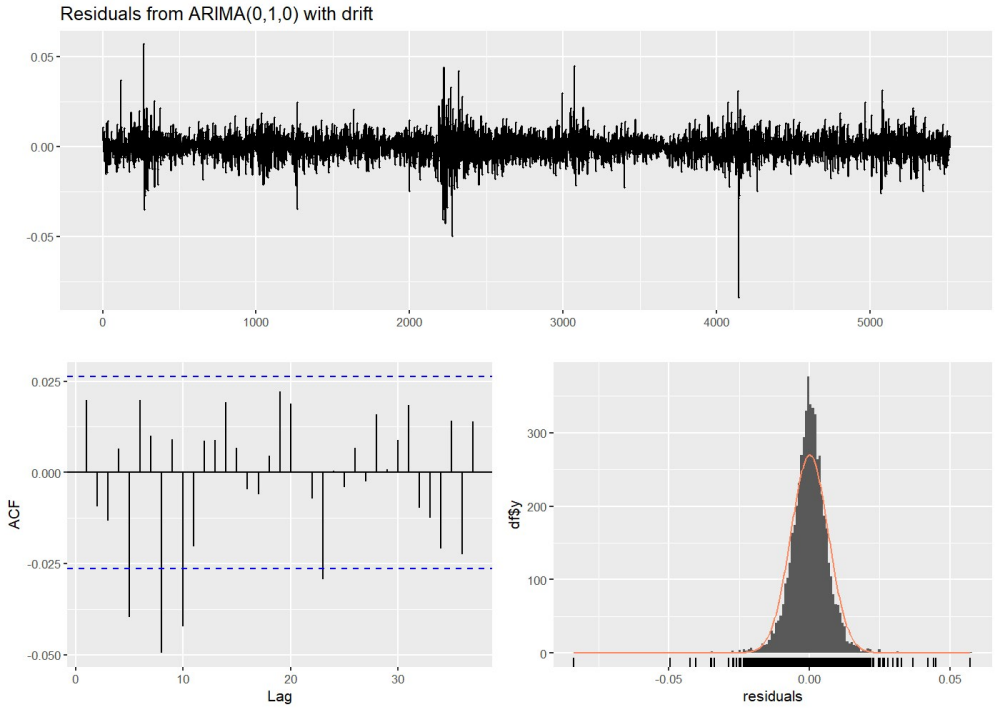


Figure 5: *Residual diagnostics of ARIMA(0,1,0) with drift, GBP/LKR (training period).*

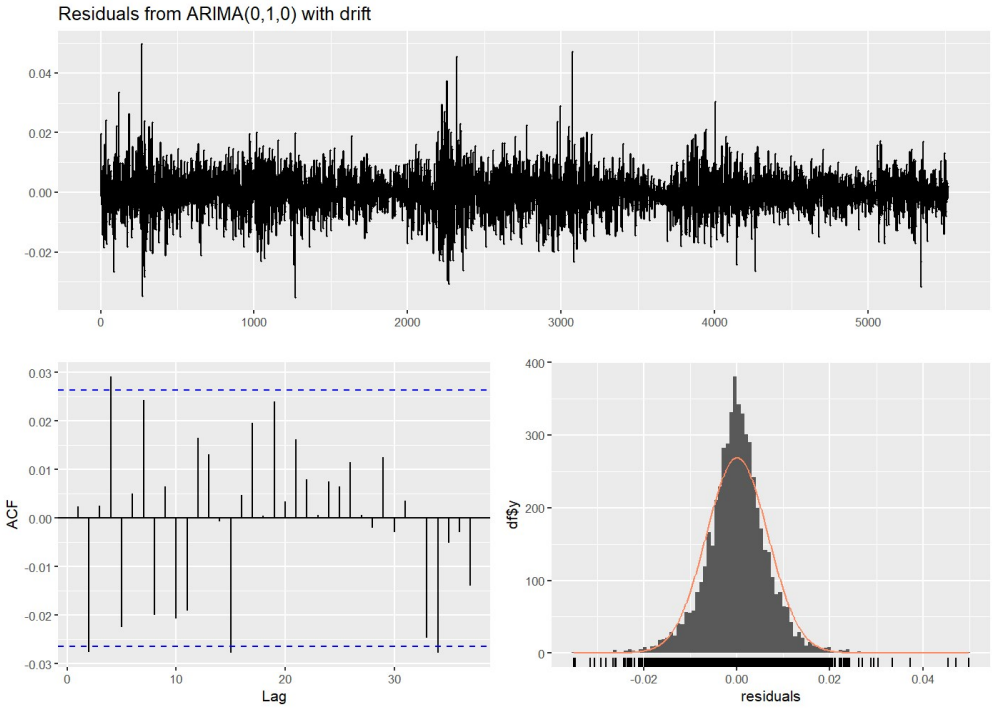


Figure 6: *Residual diagnostics of ARIMA(0,1,0) with drift, EUR/LKR (training period).*

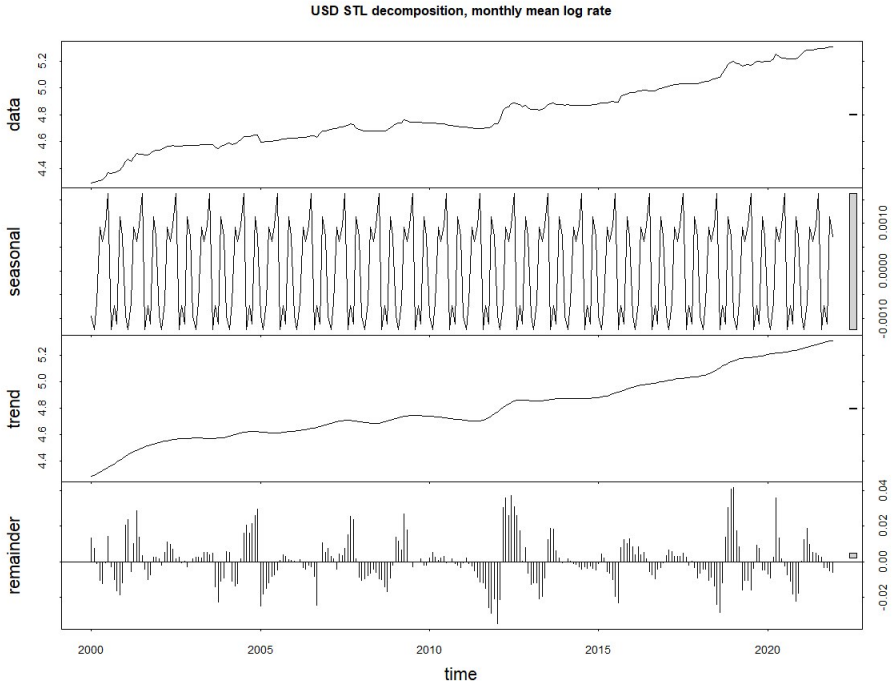


Figure 7: *STL decomposition of monthly USD/LKR (seasonality check). GBP and EUR gave similar, very small seasonal strength.*

As a supplementary illustration of why a walk-forward design was used, a single long-horizon forecast issued at the end of the training period was compared with the whole test period (static,

not walk-forward). Its test-set RMSE was 119.9 (USD), 123.7 (GBP) and 110.4 (EUR) LKR, compared with RMSE values of about 1.6–8.6 LKR for the walk-forward forecasts in Chapter 5, because the 2022 rupee float lay far outside the range of the static forecast.

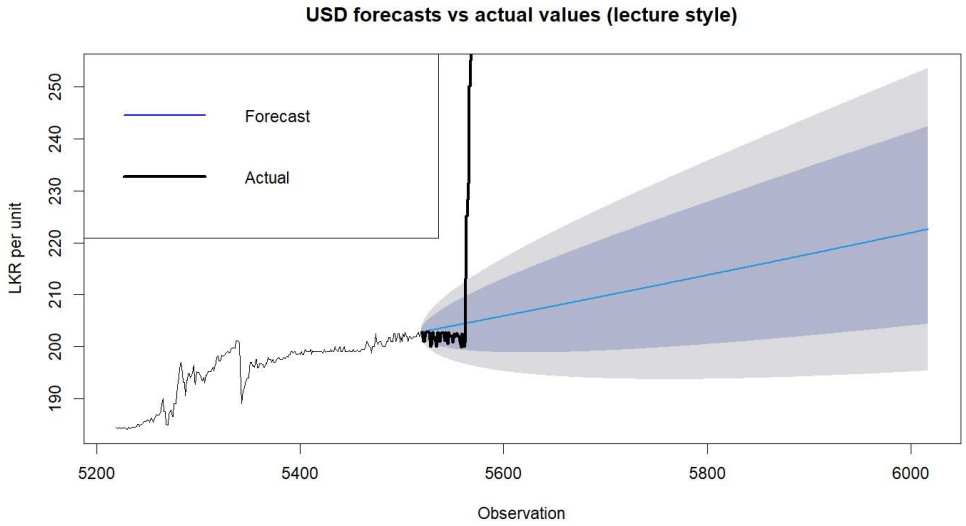


Figure 8: *Static long-horizon forecast versus actual USD/LKR (supplementary illustration; the 2022 float is far outside the forecast interval).*

Appendix E: Supplementary volatility and interval figures

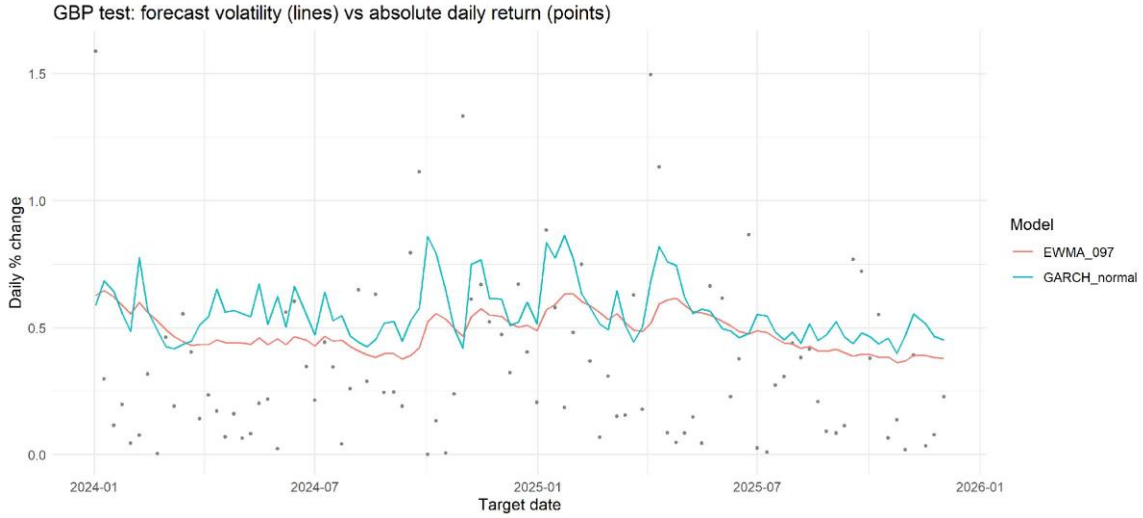


Figure 9: *GBP/LKR: forecast conditional volatility (lines) against absolute observed daily returns (points), test period.*

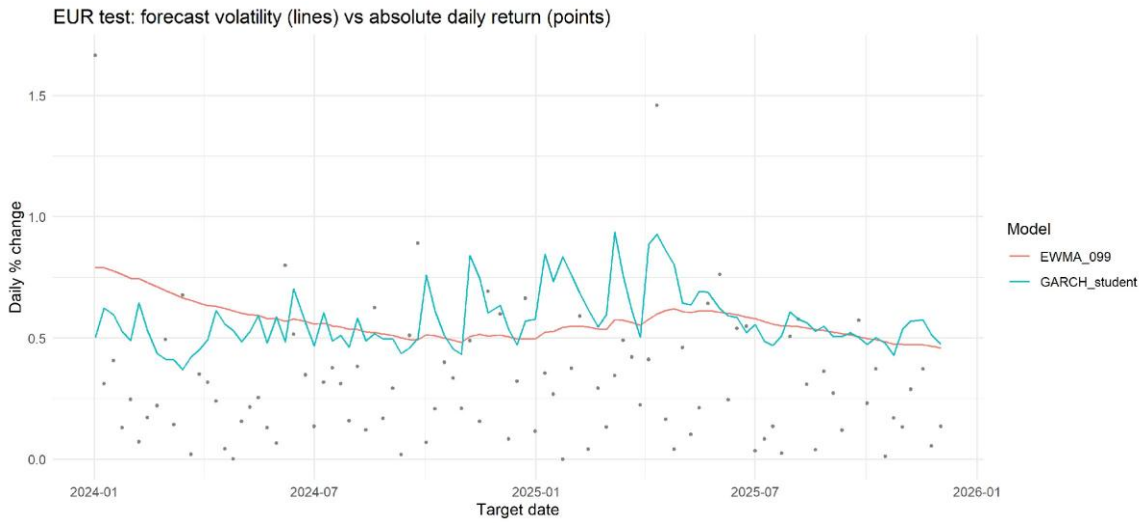


Figure 10: *EUR/LKR: forecast conditional volatility (lines) against absolute observed daily returns (points), test period.*

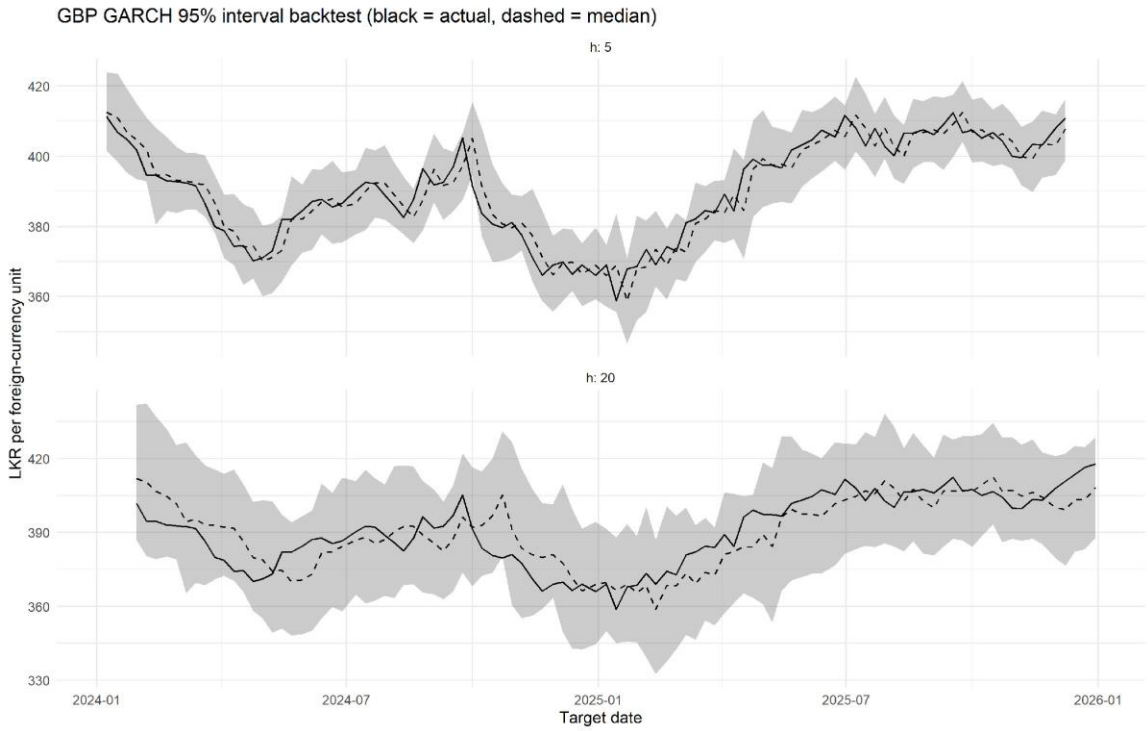


Figure 11: *GBP/LKR GARCH 95% interval backtest, test period (black = actual, dashed = median).*

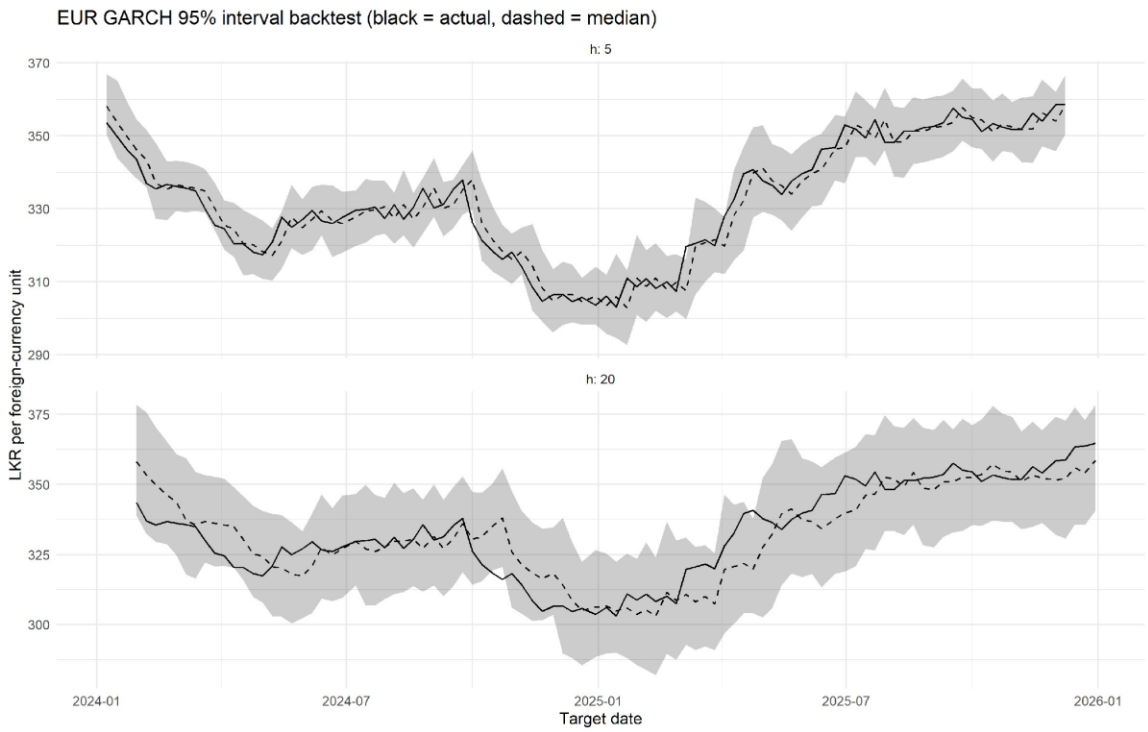


Figure 12: *EUR/LKR GARCH 95% interval backtest, test period (black = actual, dashed = median).*

Appendix F: Software environment

Table 8: *R version and main packages*

Component	Version
R	4.6.1 (2026-06-24)
forecast	9.0.2
ranger	0.18.0
xgboost	3.2.1.1
rugarch	1.5-6
vars	1.6-1
tseries	0.10-63
urca	1.3-4
lmtest	0.9-40
strucchange	1.6-0
dplyr / readr / ggplot2	1.2.1 / 2.2.0 / 4.0.3
writexl	2.0.1

Source: 11_sessionInfo.txt. Base random seed: 2026.

Appendix G: R code

https://github.com/bimsaraoshada/Time_Series_Project.git

Appendix H: Complete test forecast panels, all model families

Test forecasts of every evaluated family (top row: five-day horizon; bottom row: twenty-day horizon). Black line = actual rate. Zoom in for detail.

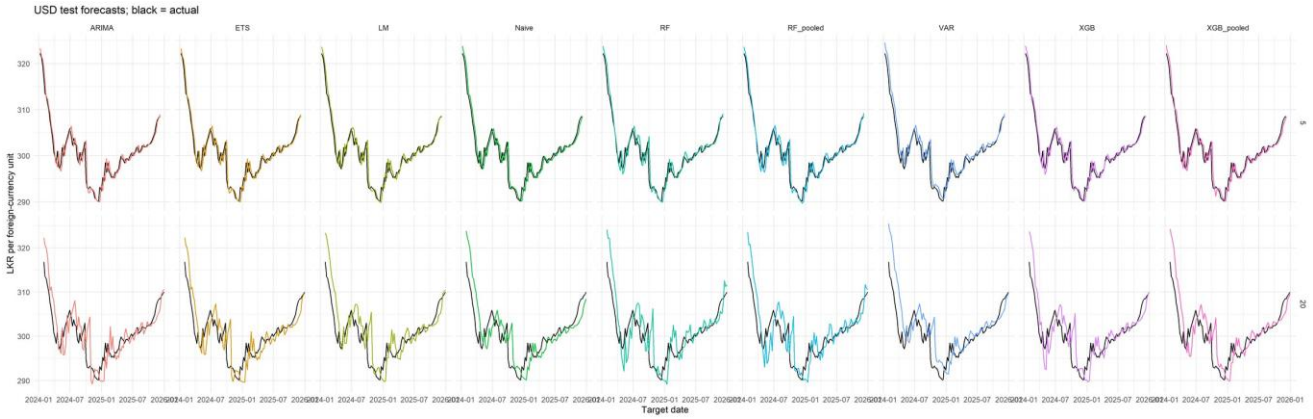


Figure 13: *USD/LKR test forecasts by model family (2024–2025).*

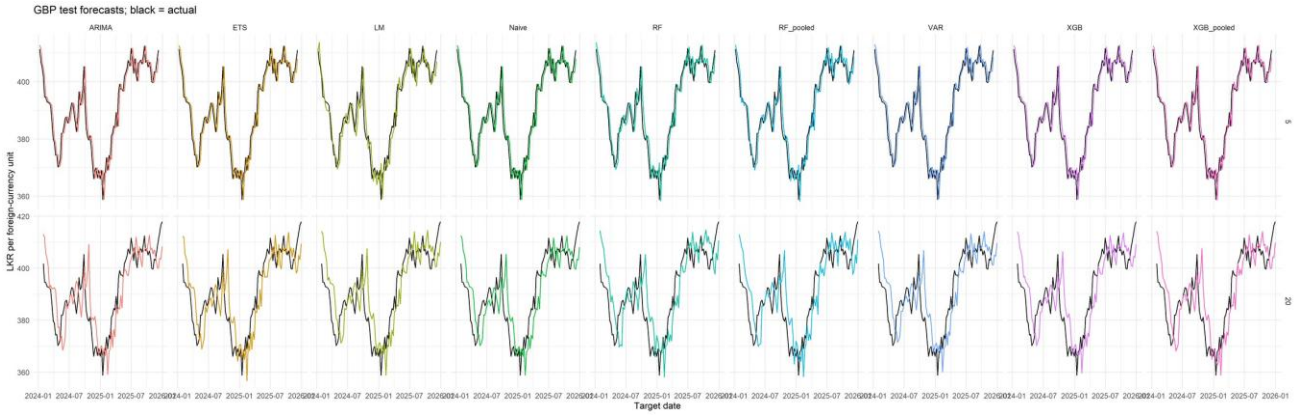


Figure 14: *GBP/LKR test forecasts by model family (2024–2025).*

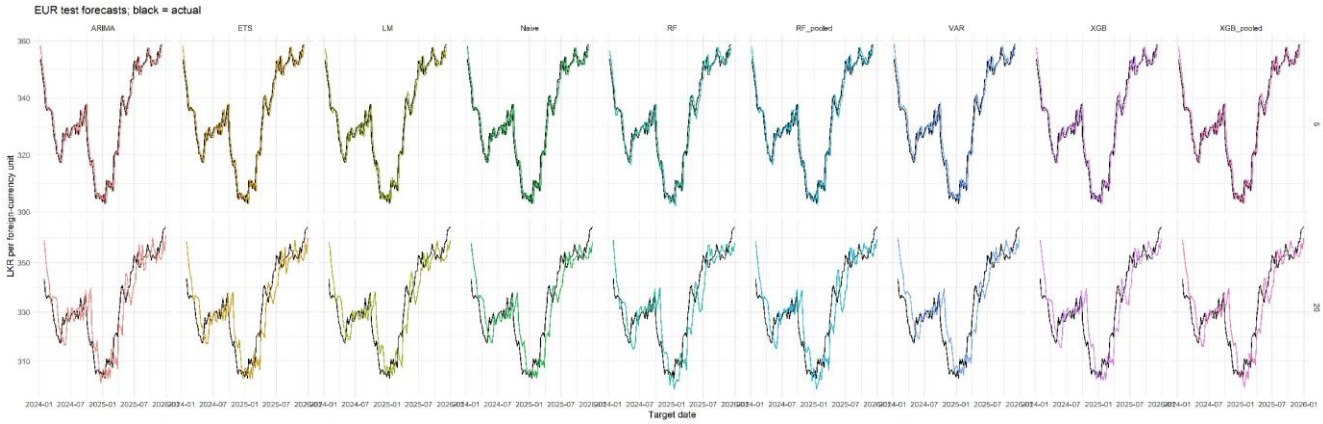


Figure 15: *EUR/LKR test forecasts by model family (2024–2025).*